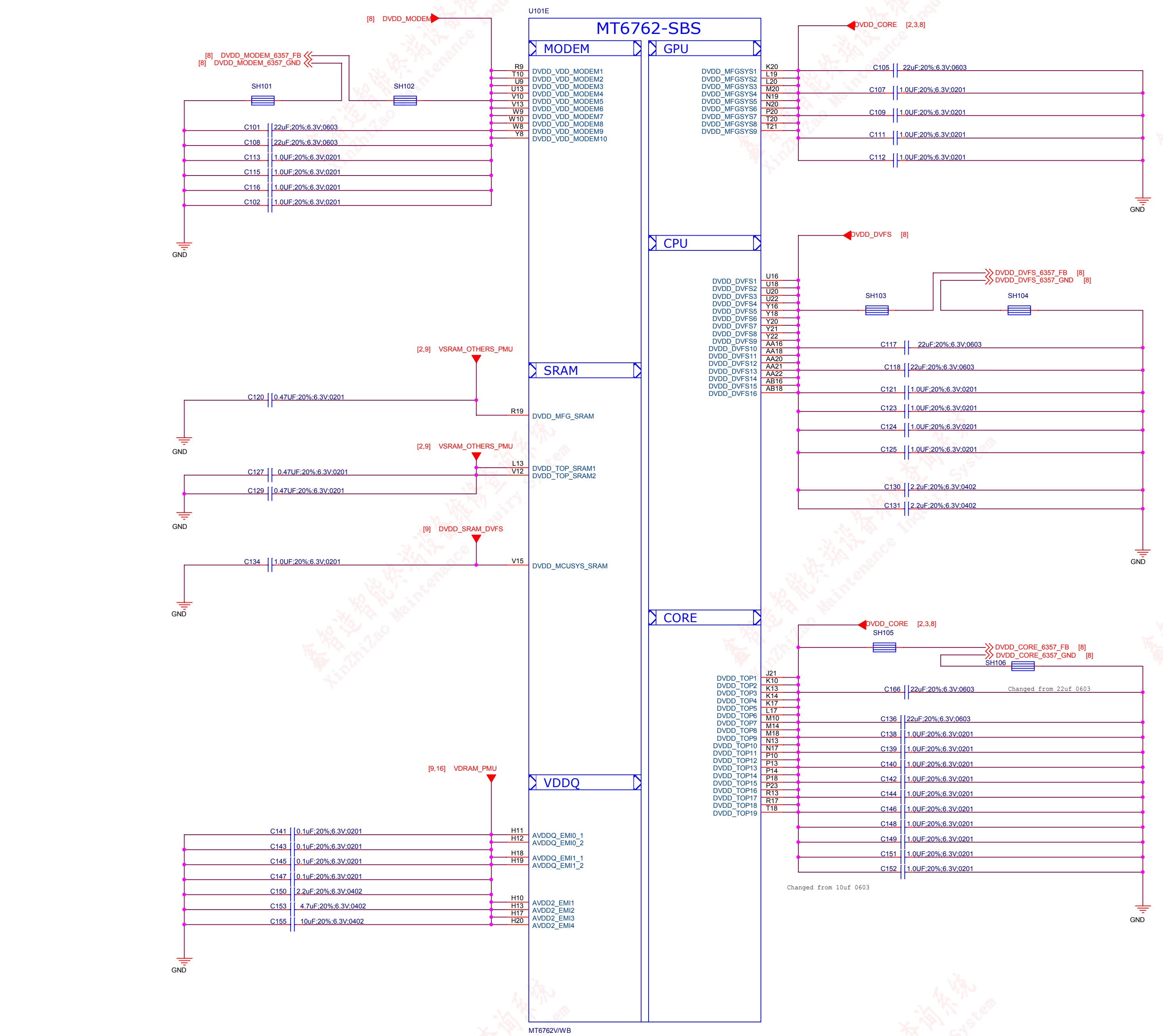
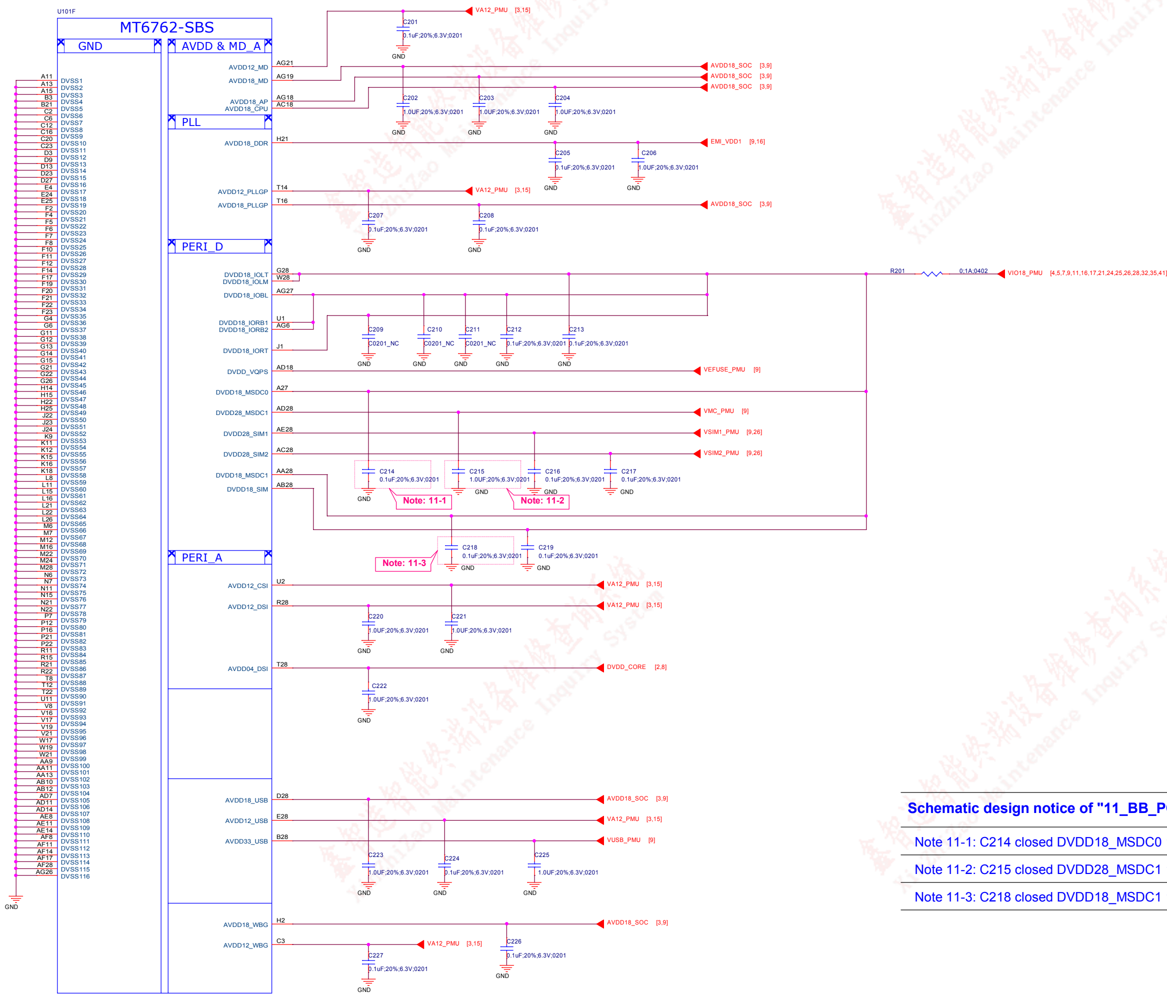
Revision History

REV	DATE	DESCRIPTION
A1	20181203	Refer to MTK DVT1.1 Reference design
A1	20181205	Change from K4 Sch 1. Change PA from 87318 to 87319 2.Change Charger IC from SM5414 to ETA6793 3.Change P-L sensor from STK3331 to Spring board STK3332 4. Delete LCM I2C switch 5. Delete USB switch 6.Delete KEY Pressed force download circuit 7.Change CAM I2C connection 8.Change BAT CONN

Title	00_History	REV: V10	
DOCUMENT NO.:	S96116-1-13_MB_20190409-1317	Size A2	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 1	of 53



Title	01_BB_POWER_PDN	REV: V10
DOCUMENT NO.:	Design Name	Size A2
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
Date: Monday, April 08, 2019	Sheet 2 of 53	



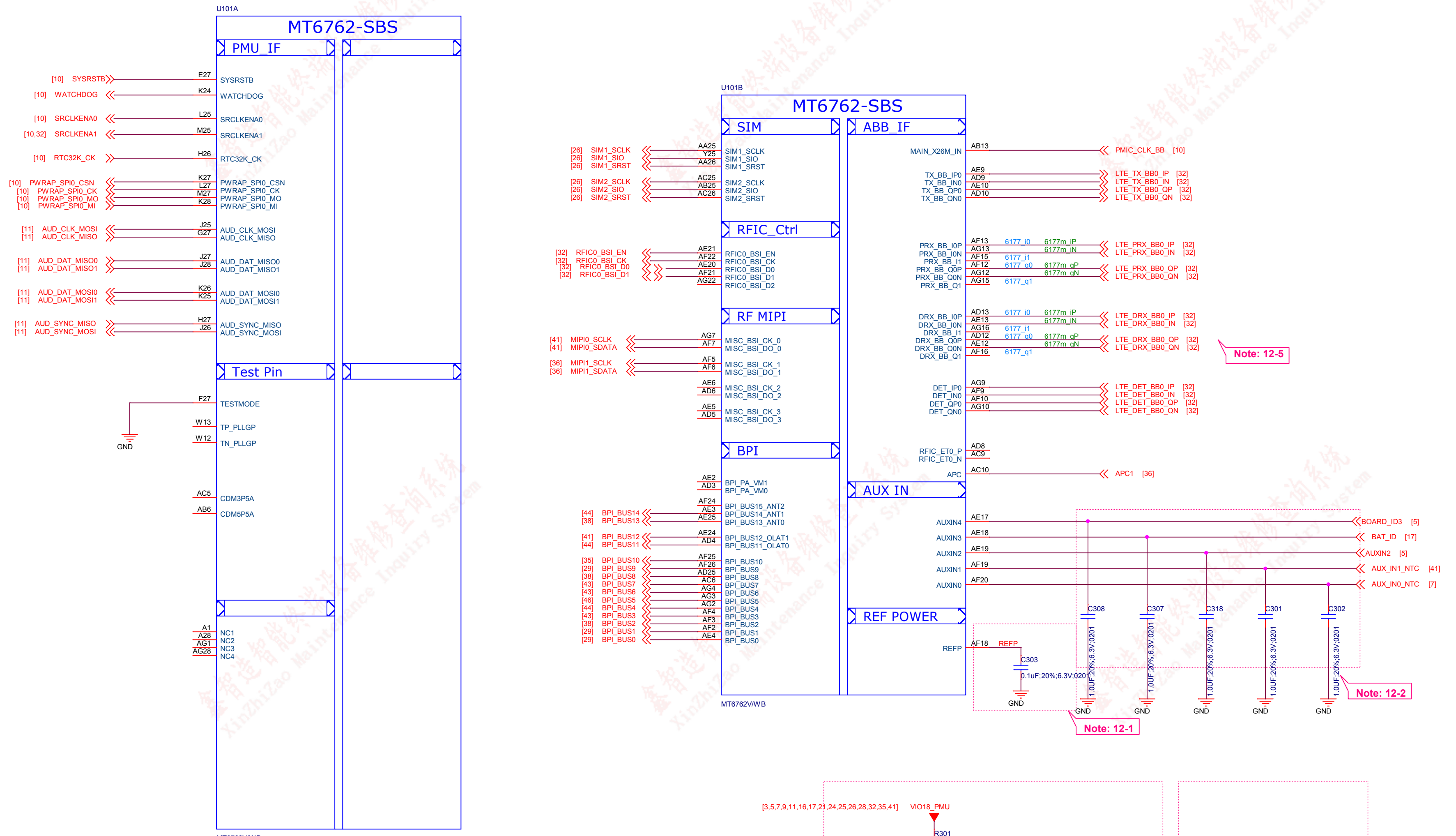
Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C214 closed DVDD18_MSDC0 150mil

Note 11-2: C215 closed DVDD28_MSDC1 150mil

Note 11-3: C218 closed DVDD18_MSDC1 150mil

Title	02_BB_POWER_IO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	3 of 53



Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

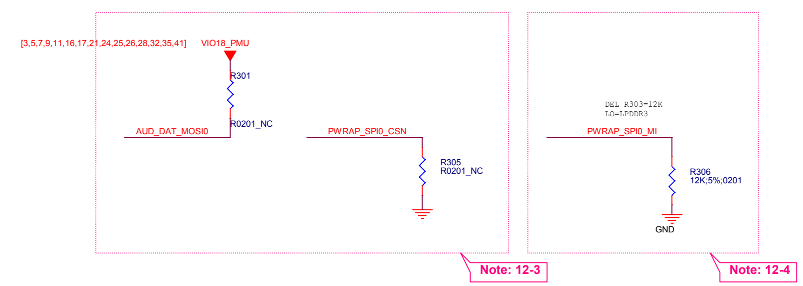
Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3
LO (by ext. PD)	LO	SPI0+EINT8	N/A
LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

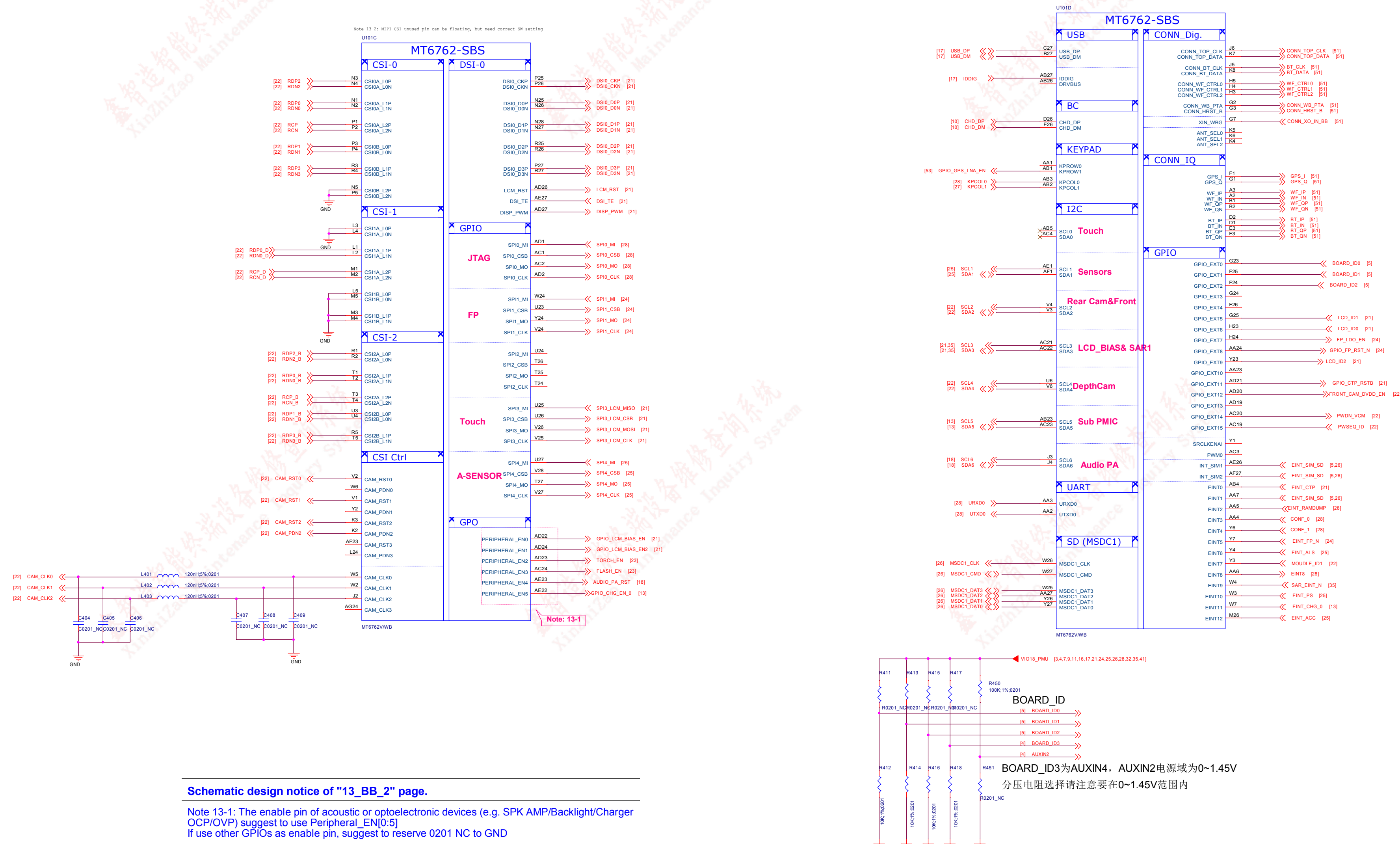
Note 12-4: PWRAP_SPI0_MI is DDR type feature in bootstrap

PWRAP_SPI0_MI	Booting interface	
default=PU	DDR	MSDC0 pin mux
LO (by ext. PD)	LPDDR3	follow LP3 Ref SCH.
HI	LPDDR4X	follow LP4X Ref SCH.



Note 12-5: Please set unused IQ pins in NC

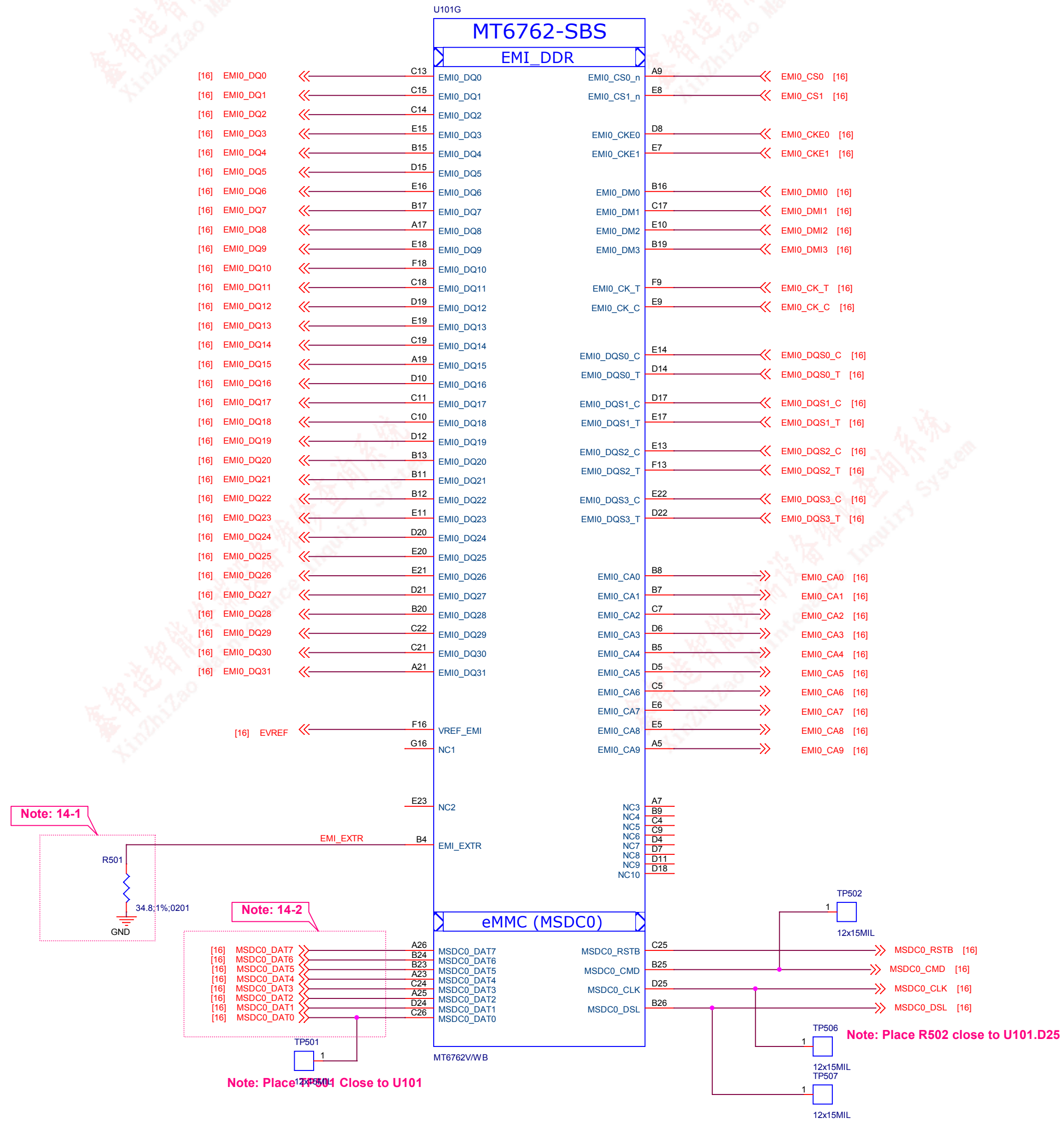
Title	03_BB_1_RF&SIM_IF	REV: V10
DOCUMENT NO.:	Design Name	Size A2
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
Date: Monday, April 08, 2019	Sheet 4 of 53	



Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

Title	04_BB_2_MIPI&GPIO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	5 of 53



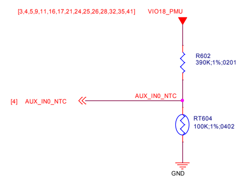
Schematic design notice of "14_BB_3" page.

Note 14-1: R501 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux

Title	05_BB_3_EMMC		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 6 of 53	

The pull up ressister need to be 1% accuracy
NTC=100K



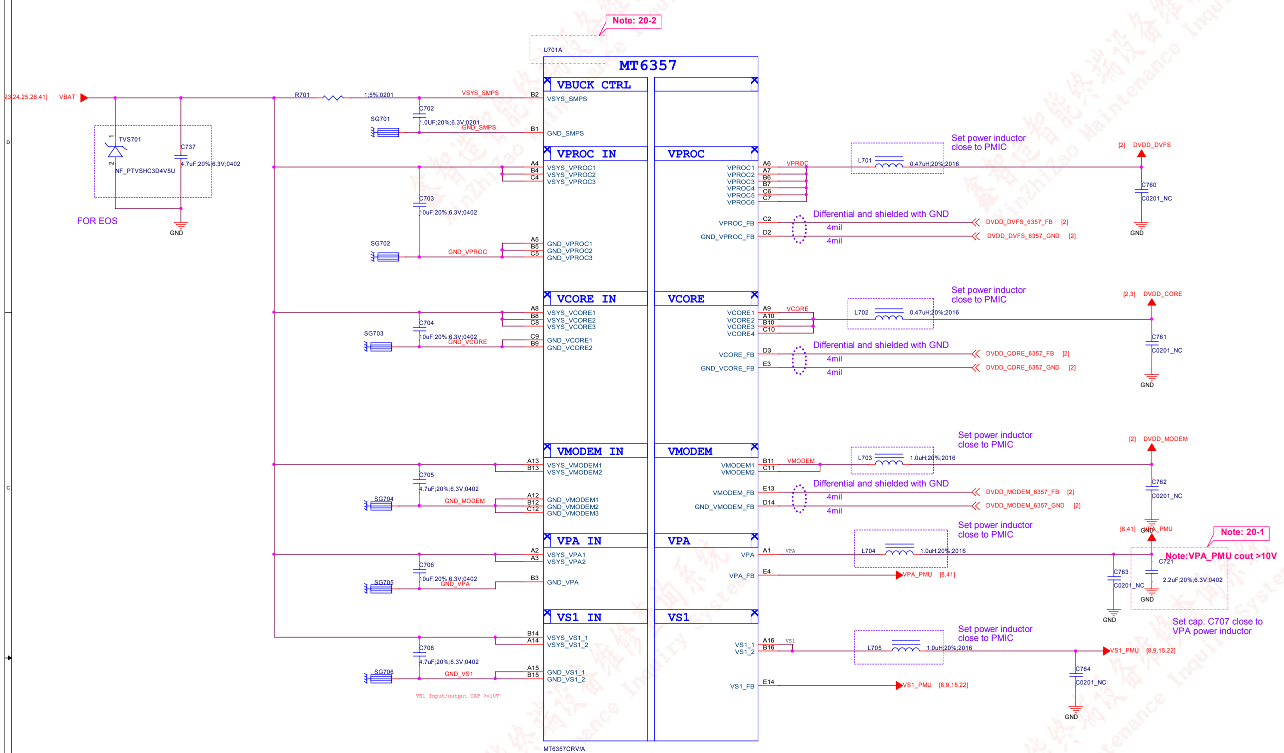
Thermistor to sense CHARGER temperature

Huawei NOTE:1. NTC606 must keep a distance about 3~5 mm away from Charger IC and far from other heat sources 5~10 mm at least.

Thermistor to sense AP temperature

1. RT604must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

Title	06_BB_AUXADC_Thermal		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 7 of 53	



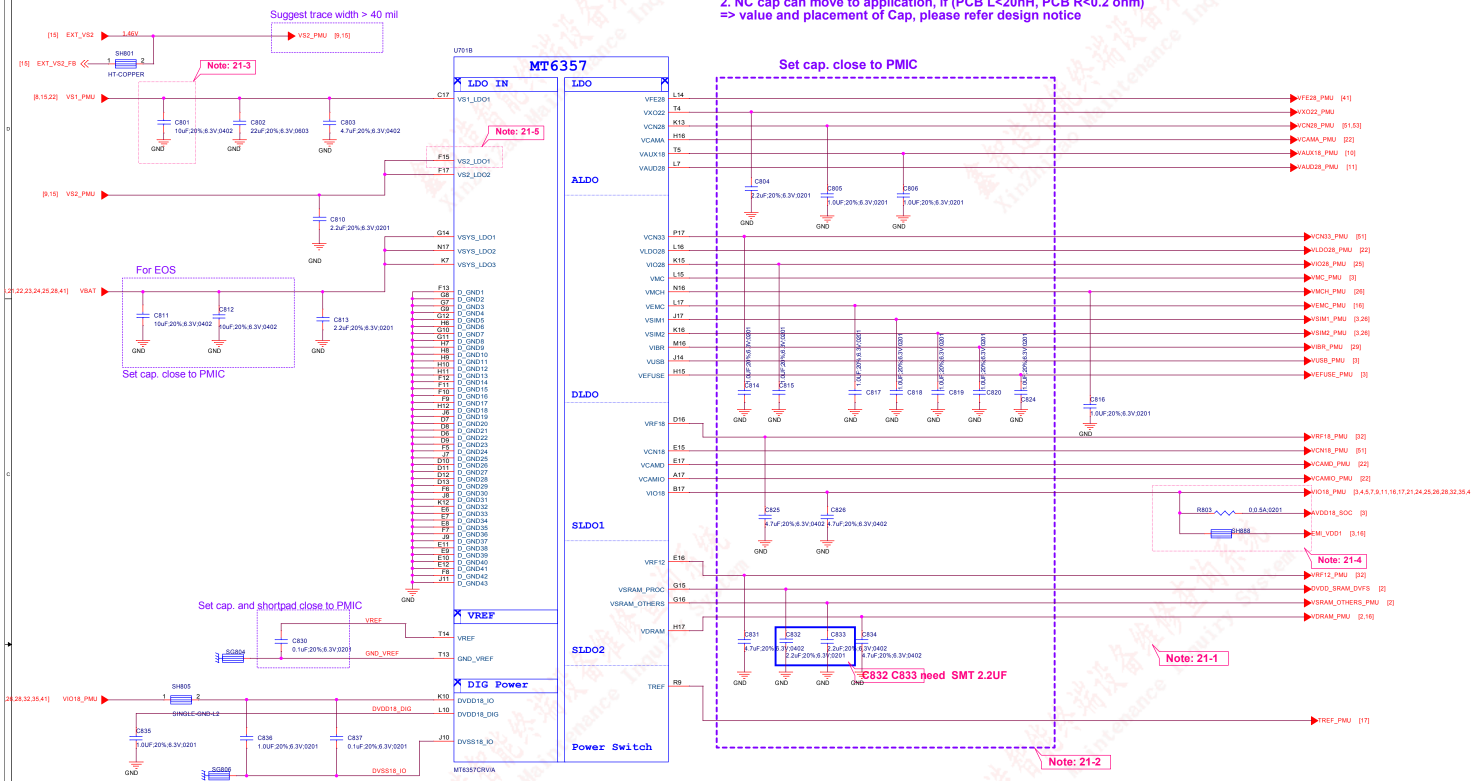
Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C707, please choose 0402 size

Note 20-2: For MT6765/62/61 platform, please only use MT6357CRV

Title	07_POWER_MT6357_Buck	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 8	of 53

1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2 ohm)
=> value and placement of Cap, please refer design notice



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

Note 21-3: Ext Buck BOM option

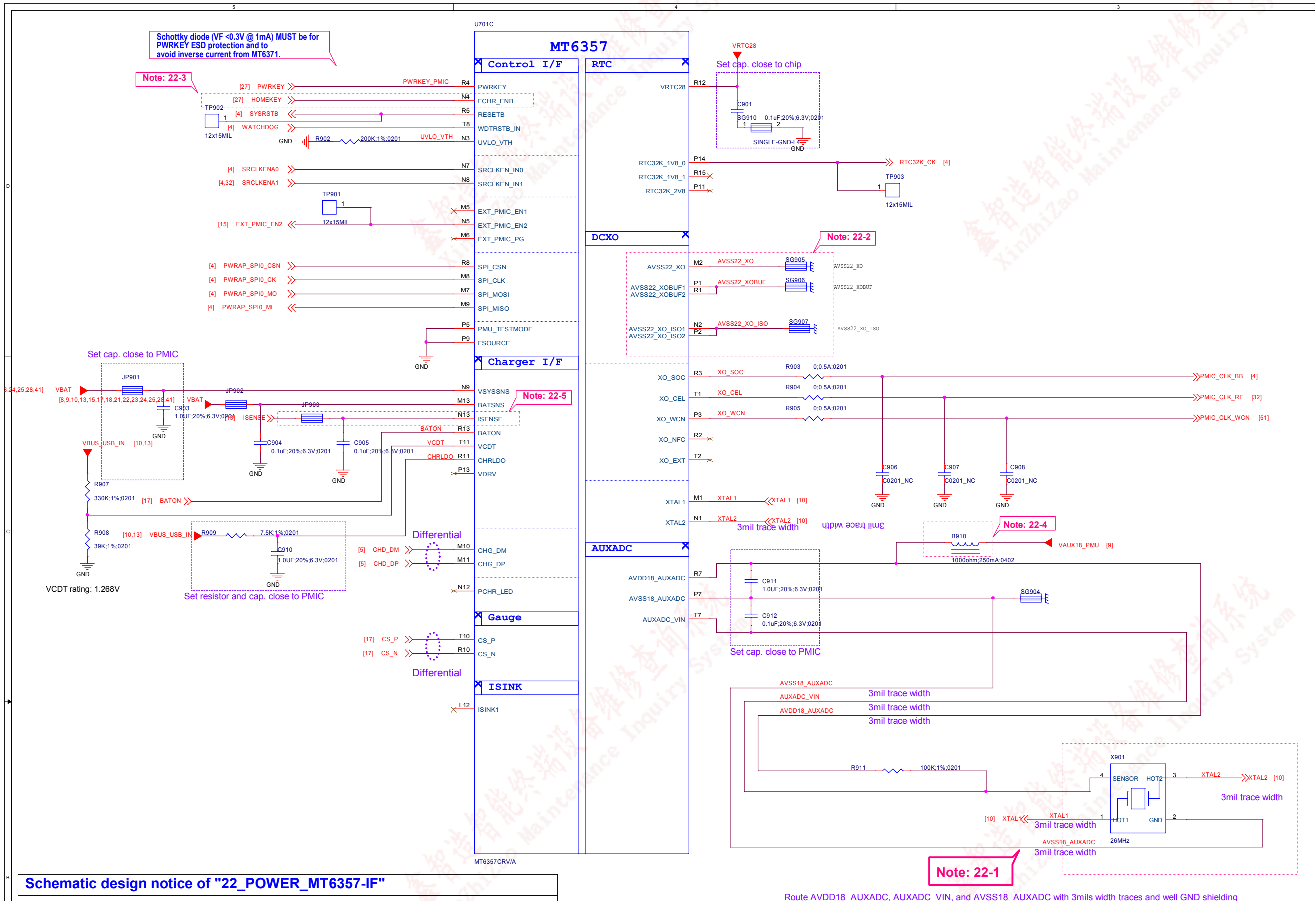
	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C801	10uF	22uF

Note 21-4: Please set R803 and R803 close to C826, making star connection among VIO18_PMU, AVDD18_SOC, and EMI_VDD1 near to LDO cap. C826

Please also refer to MT6357 design notice for further detail design information

Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) >= 1.3 V

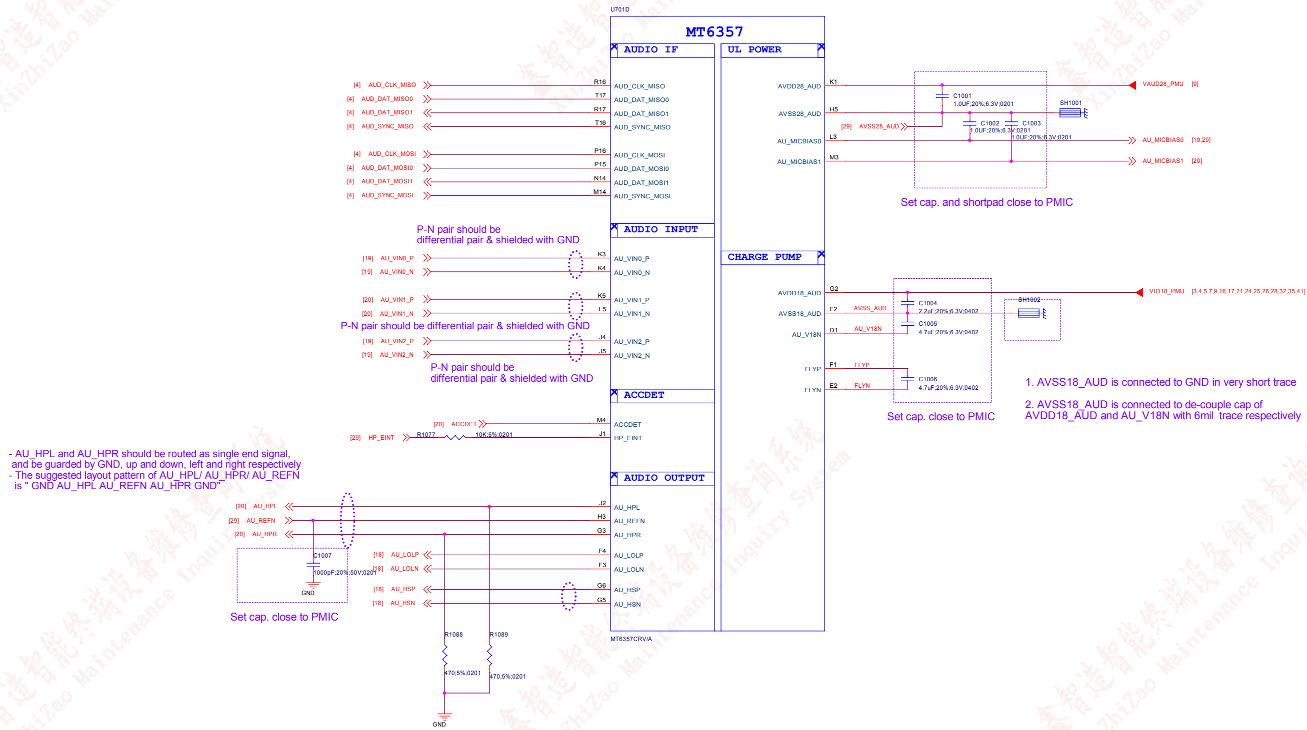
Title	08_POWER_MT6357_LDO	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 9	of 53



Schematic design notice of "22_POWER_MT6357-IF"

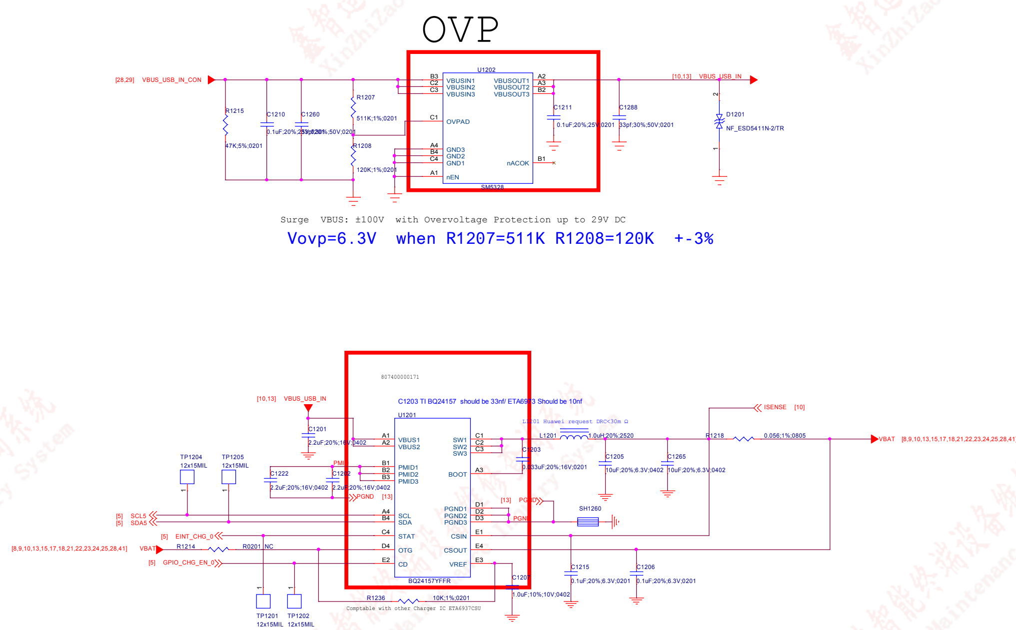
Note 22-1:	Please implement 2520 & 2016 Size TMS PCB co-layout. Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide
Note 22-2:	1. Please Connect P1 and R1 ball first and then to GND 2. Please Connect P2 and N2 ball first and then to GND 3. Please connect DCXO GND to main GND by independent L1-2 GND via.; DO NOT connect it through L1 GND
Note 22-3:	Let floating if disable HOMEKEY function
Note 22-4:	Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R8101 can use 0 ohm to replace BEAD. Please route VAUX18_PMU with well-ground sheilding if using 0 ohm to replace BEAD for AVDD18_AUXADC
Note 22-5:	Please connect to battery connector

Title	09_POWER_MT6357_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	10 of 53



Title	10_POWER_MT6357_Audio	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 11	of 53

Title	11_POWER_SubPMIC-General	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 12	of 53



Schematic design notice of "26_POWER_SubPMIC-Charger + PP" page.

Note 26-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

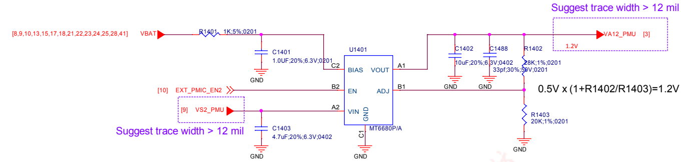
Title	12_POWER_SubPMIC-Charger + PP
DOCUMENT NO.:	Design Name
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019	Sheet 13 of 53

WINGTECH

Title 13_POWER_SubPMIC-HV powers	REV: V10
DOCUMENT NO.: Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI
WINGTECH	
Date: Monday, April 08, 2019	Sheet 14 of 53

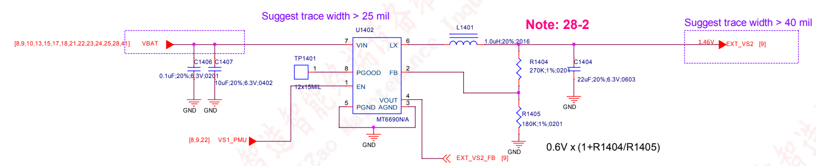
LDO for VA12

Note: 28-1



Ext. Bulk for VS2

Note: 28-2

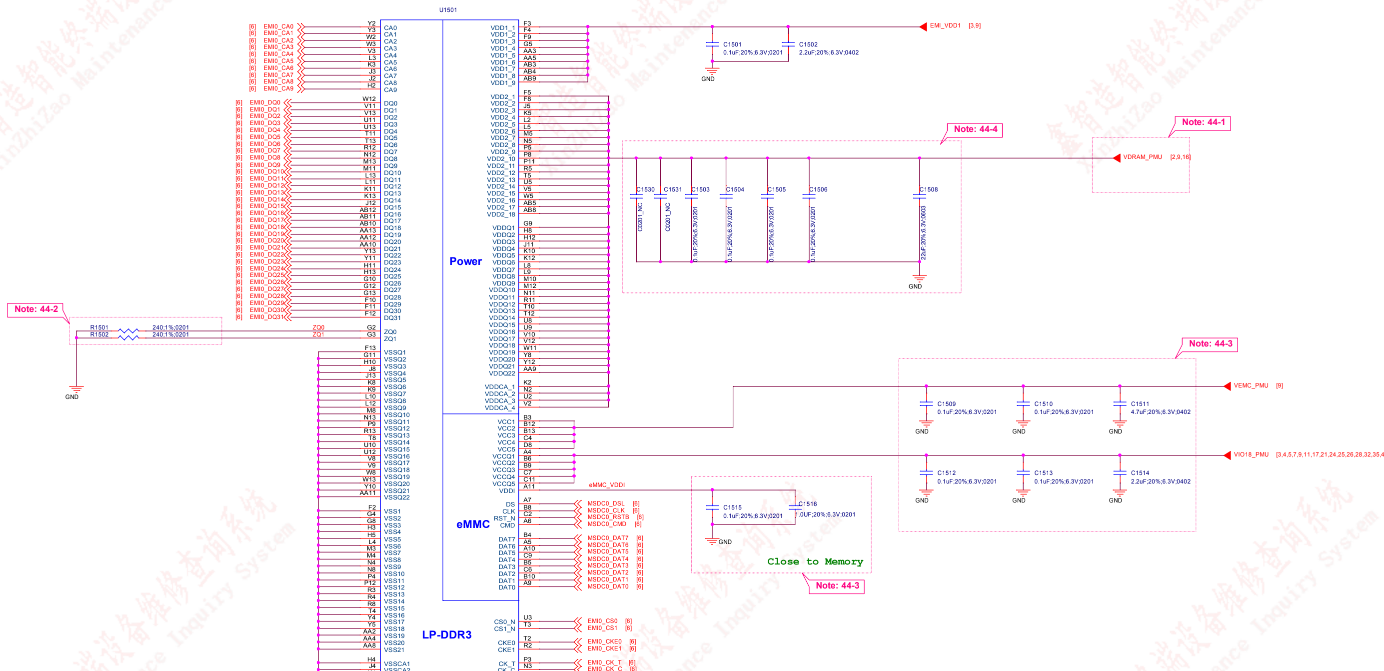


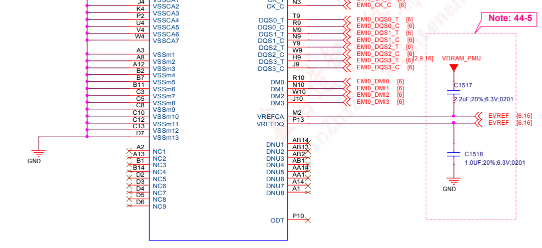
NOTE: Do not use VCAMD 1.3/1.5/1.8 when VS2 BUCK applied

Schematic design notice of "28_POWER_ThirdParty-Power"	
Note 28-1:	VA12 Layout placement please close to AP
Note 28-2:	VS2 Buck Layout placement please close to PMIC MT6357
Note 28-3:	VCN33 LDO Layout placement please close to MT6631

Title	14_POWER_ThirdParty_Powers	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	15 of 53

LPDDR3





Schematic design notice of "44_Memory_eMMC_LPDDR3"

Note 44-1: Please refer to power supply related page select VDRAM1 output voltage properly for LPDDR3

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to GND

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

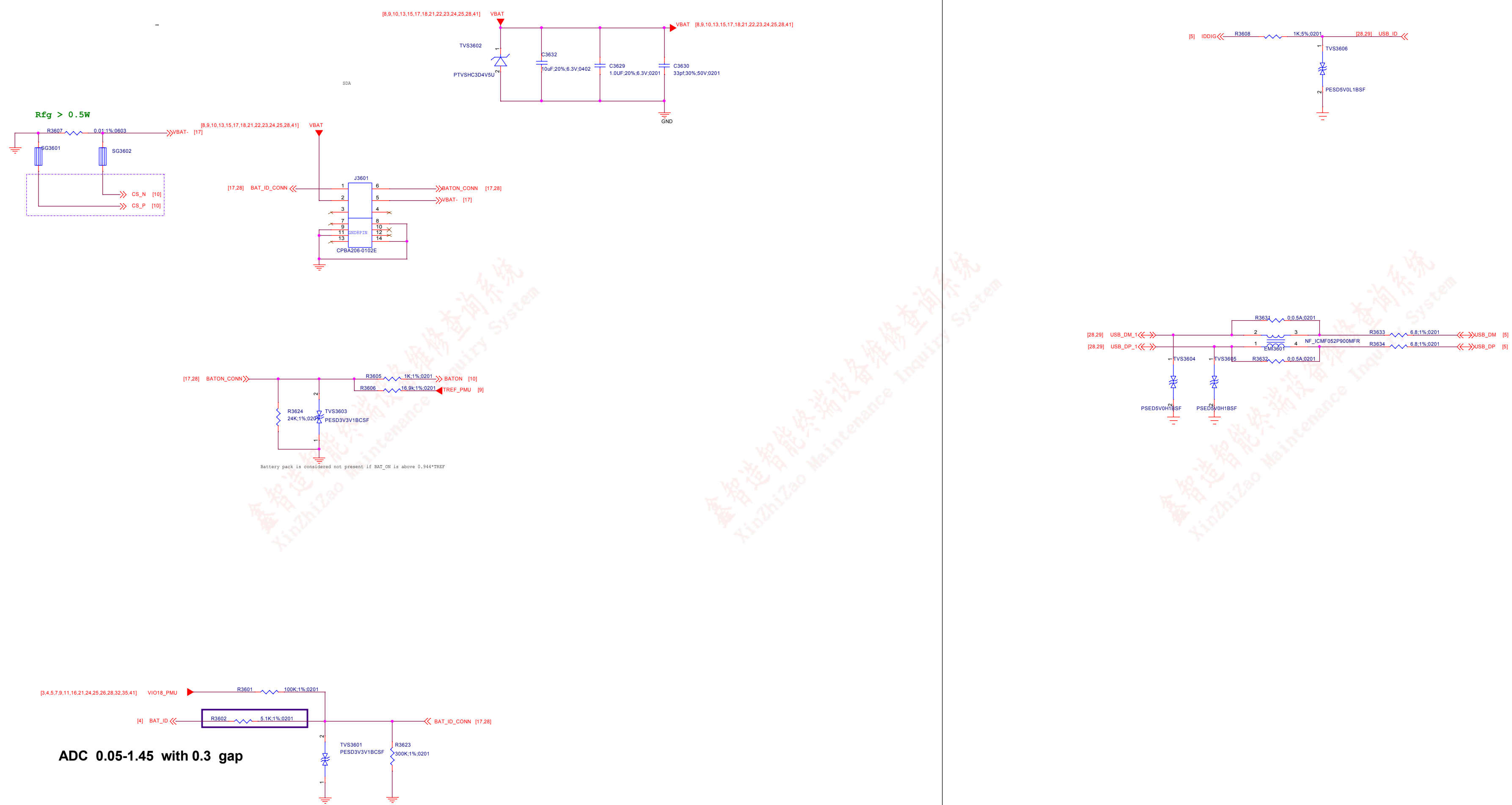
Note 44-4: VDD2 VDDQ VDDCA decoupling cap: closed to DRAM ball.
For other cap for PMIC [>10uF, at PMIC page]:
please also refer to MMD and layout guide for placement.

Note 44-5: Please check MT6765, MT6762 and MT6761's capacitor value.

Project	C1517	C1518
MT6765	2.2uF	1uF
MT6762	2.2uF	1uF
MT6761	0.1uF	0.1uF

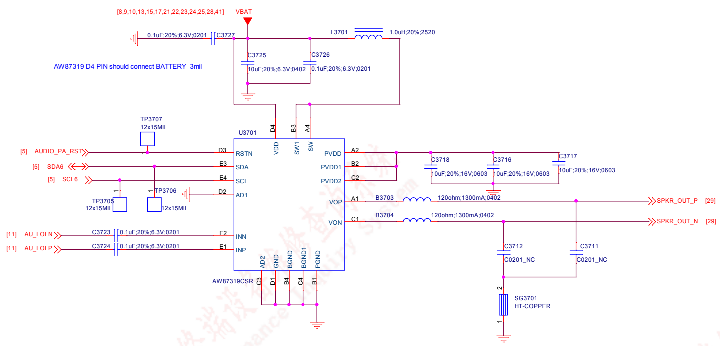
Title	15_Memory_eMMC_LPDDR3	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	YANGSHAOYUN
Date:	Monday, April 08, 2019	Sheet	16 of 53

BATTERY CONNECTOR

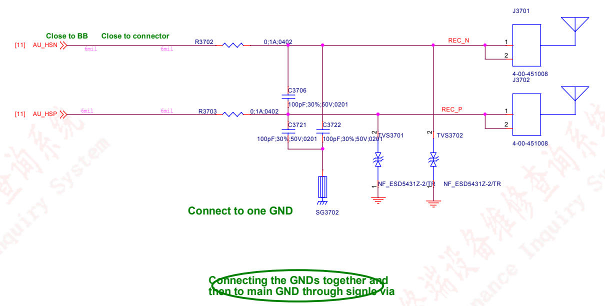


Title	36_Battery/USB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 17 of 53	

SPK

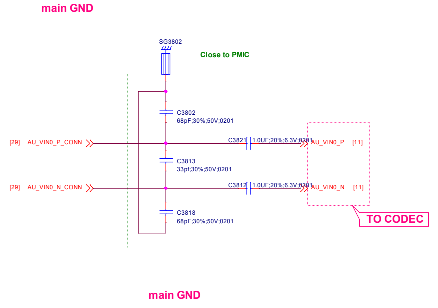


Receiver

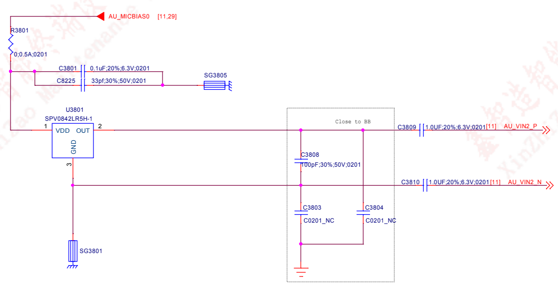


Title	37_SPK/Receiver/Vibrator	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	18 of 53

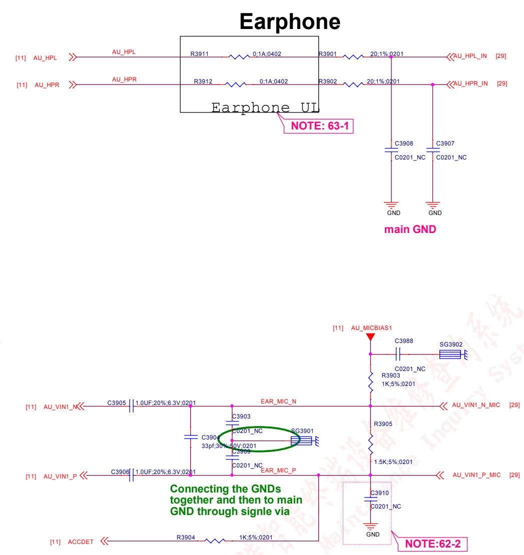
MAIN MIC



TOP MIC



Title	38_MIC	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	19 of 53



NOTE:62-1 R3911 R3912, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Title	39_Earphone	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
Date: Monday, April 08, 2019	Sheet 20 of 53	

WINGTECH

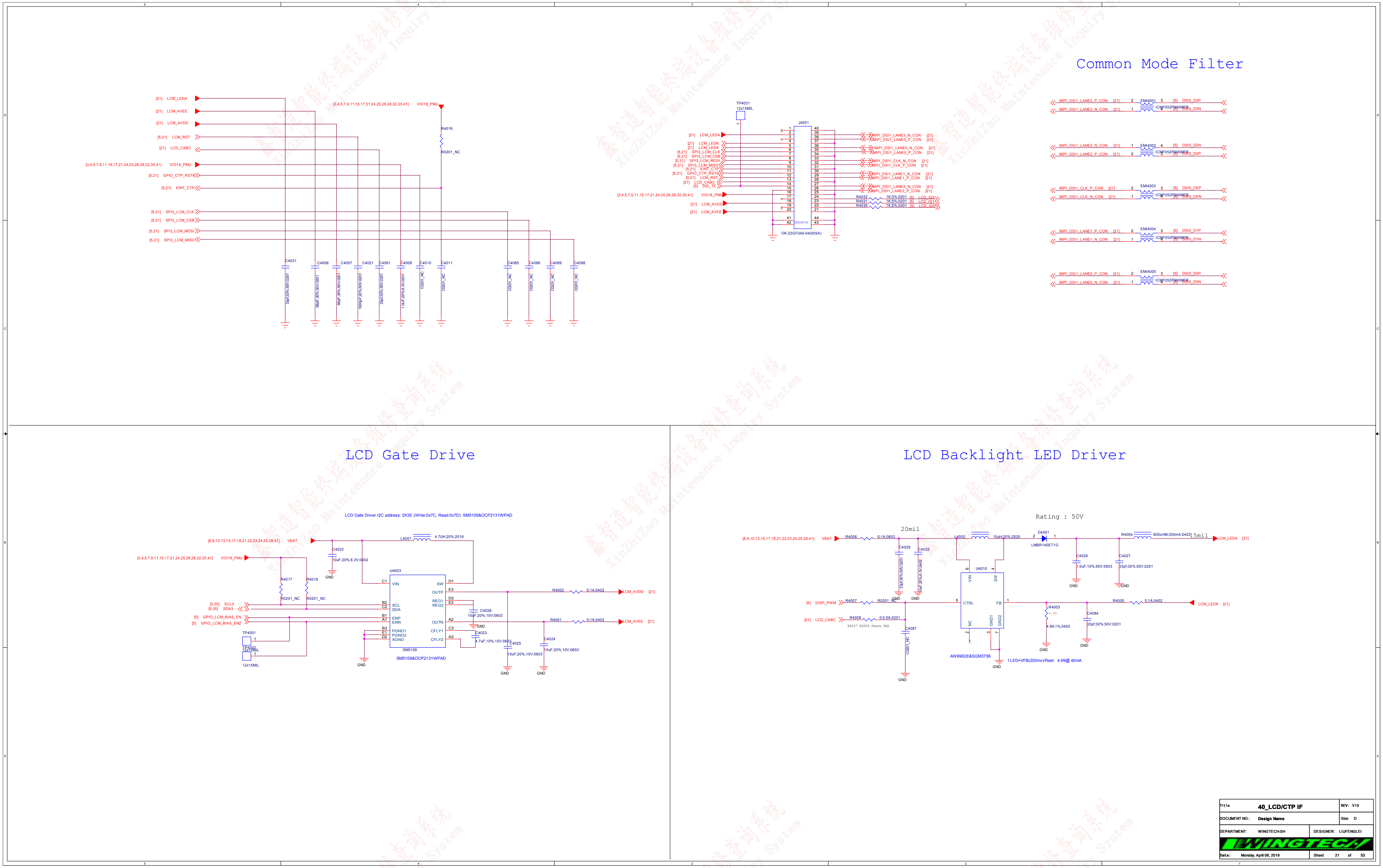
Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

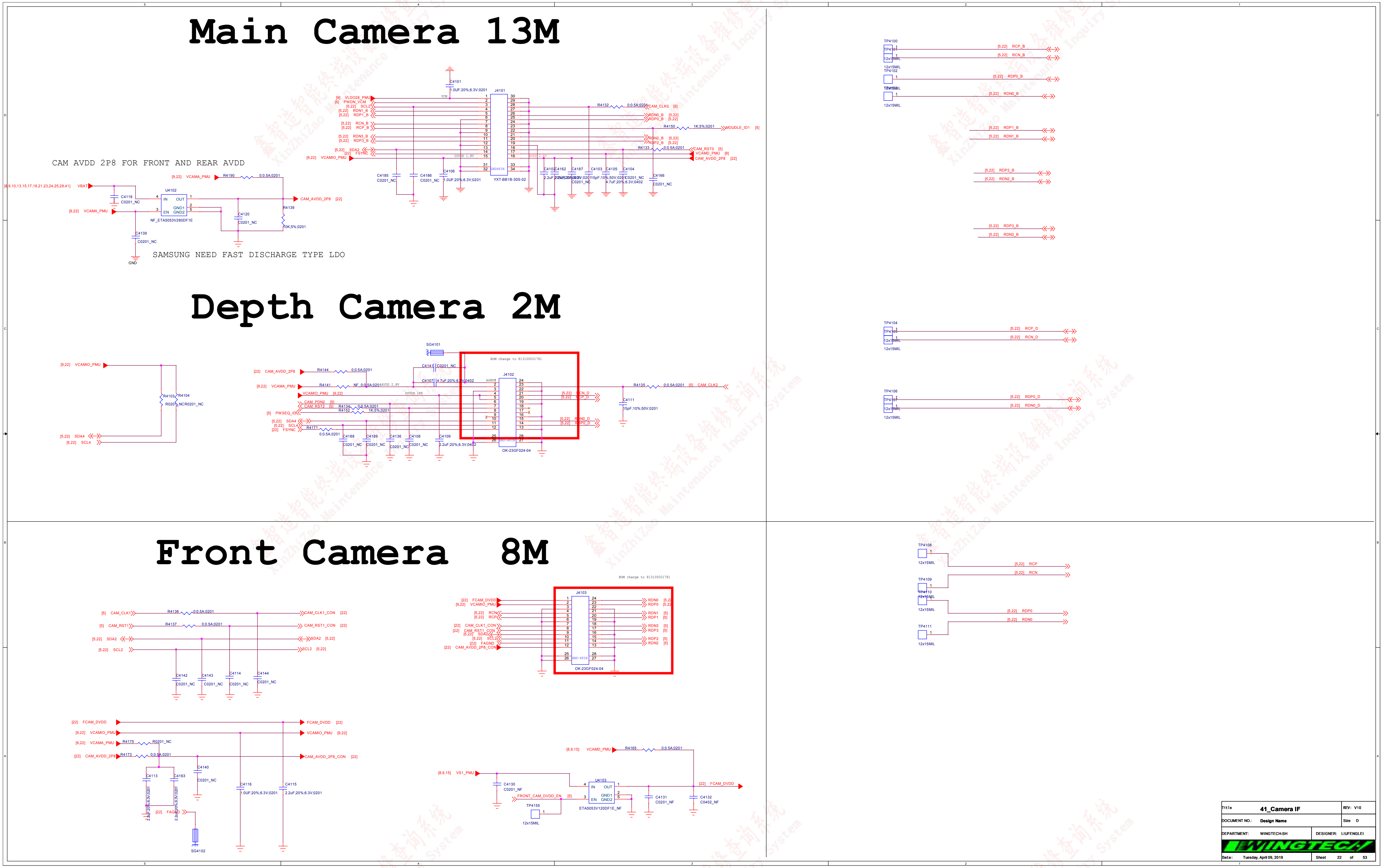
Note 62-3:

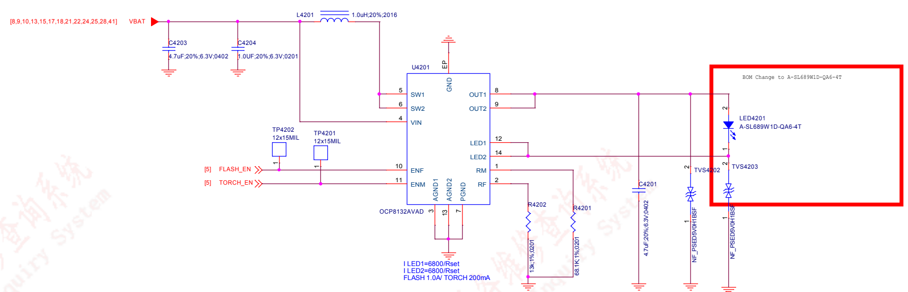
	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test;

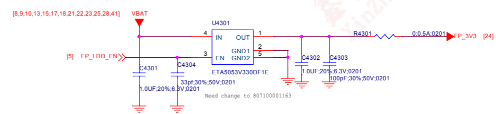


Common Mode Filter
LCD Gate Drive
LCD Backlight LED Driver
LCD Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D) SM5109&OCP2131WPAD
SM5109&OCP2131WPAD
Rating : 50V
20mil
15mil
AW9962E&SGM3756
I LED=VFB(200mv)/Rset 4.99@ 40mA
OK-23GF040-04(005A)
40_LCD/CTP IF
Design Name
WINGTECH-SH
LIUFENGLEI
WINGTECH
Monday, April 08, 2019

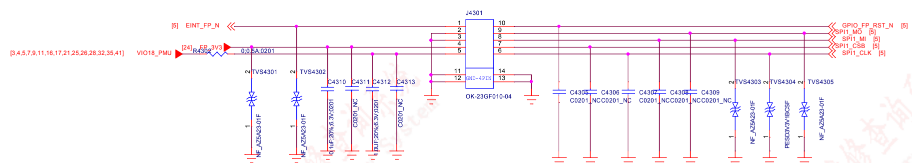

Main Camera 13M
CAM AVDD 2P8 FOR FRONT AND REAR AVDD
SAMSUNG NEED FAST DISCHARGE TYPE LDO
Depth Camera 2M
Front Camera 8M
41_Camera IF
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Tuesday, April 09, 2019
Sheet 22 of 53



Title	**42_Flash/RGB**	REV: V10
DOCUMENT NO.:	**Design Name**	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019		Sheet 23 of 53

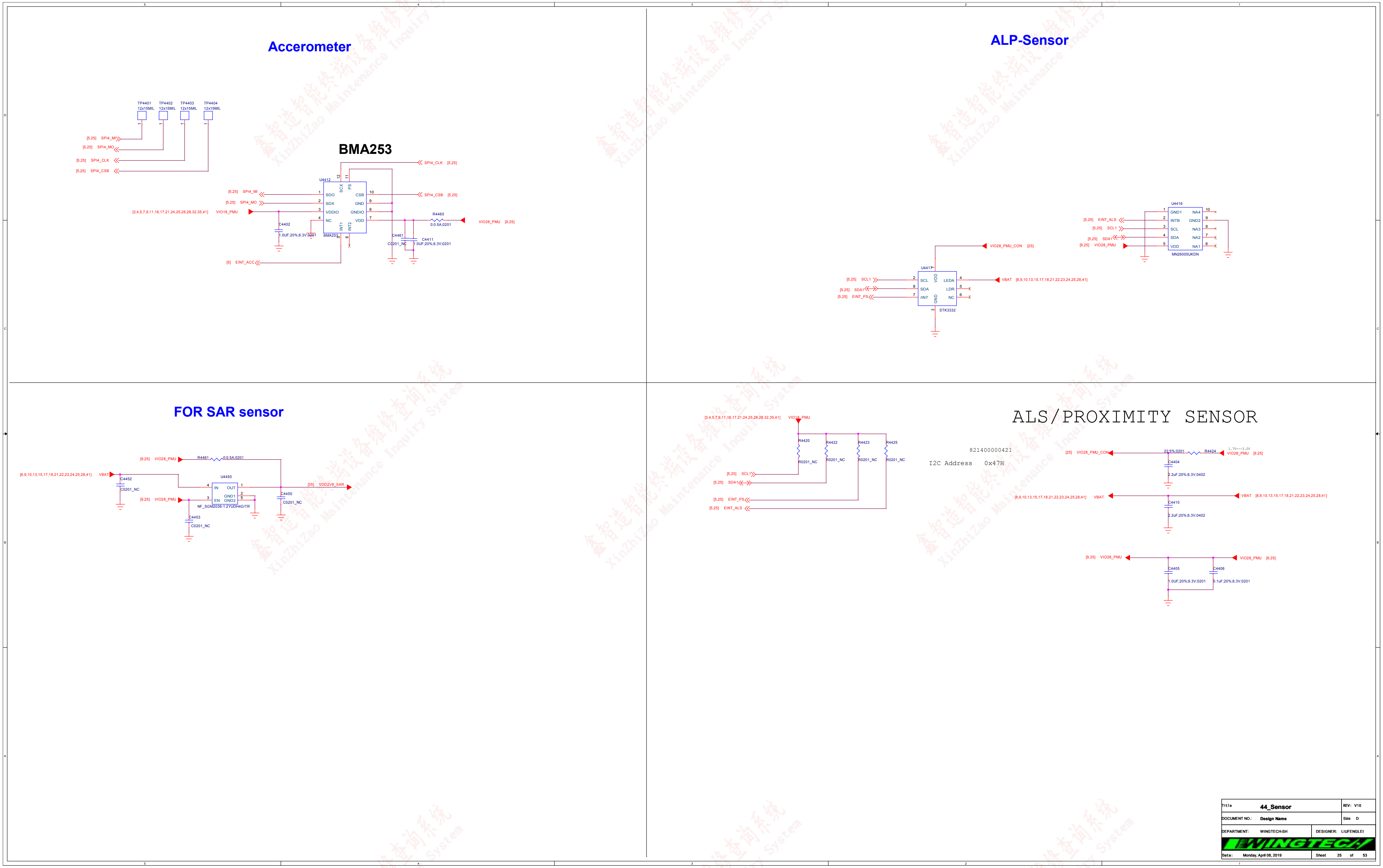


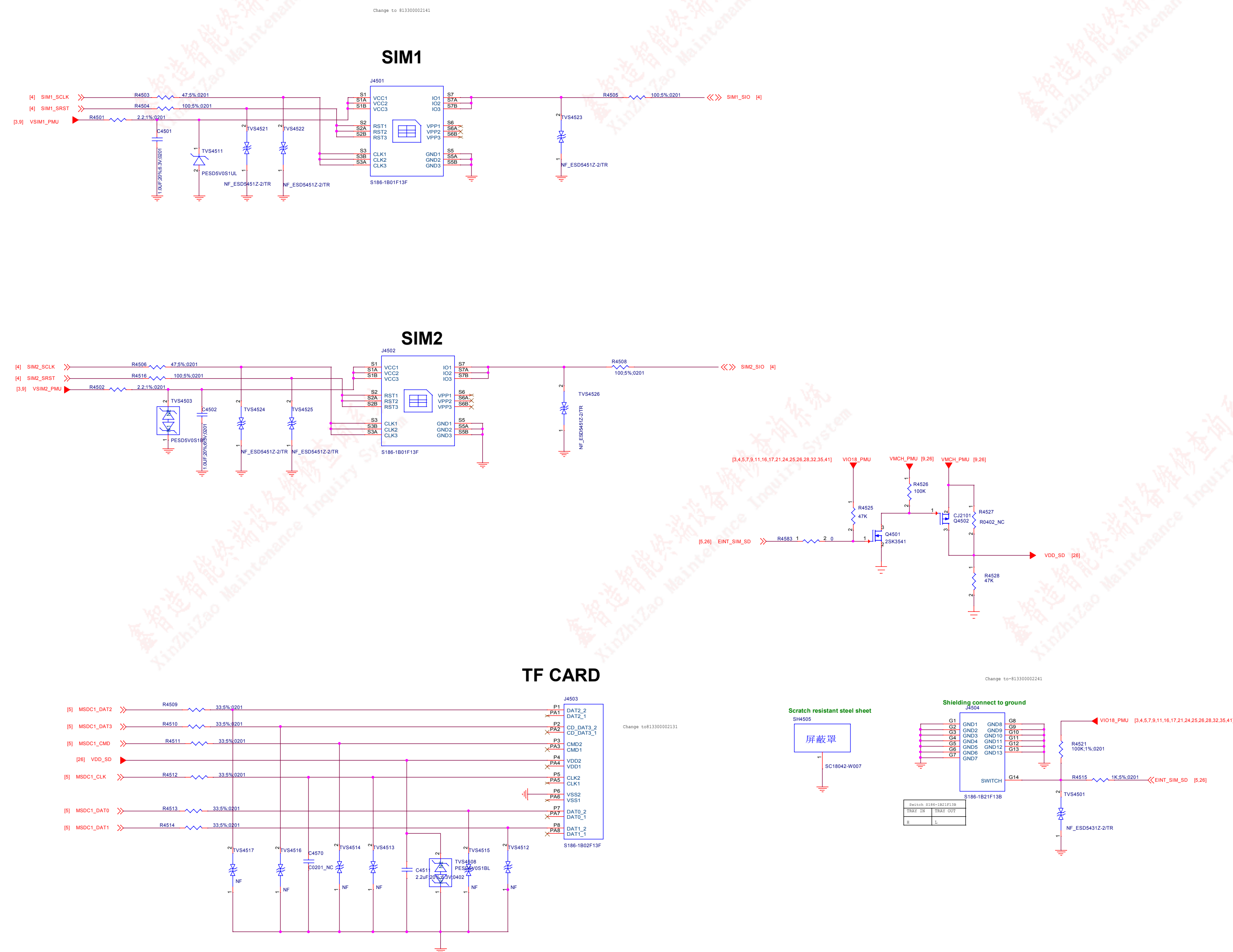
Fingerprint



Fingerprint

Title	43.Fingerprint	REV: V10	
DOCUMENT NO.:	S96116-1-13_MB_20190409-1317	Size	D
DEPARTMENT:	Wingtech-SZ	DESIGNER:	DESIGNER
WINGTECH			
Date:	Monday, April 08, 2019	Sheet	24 of 53

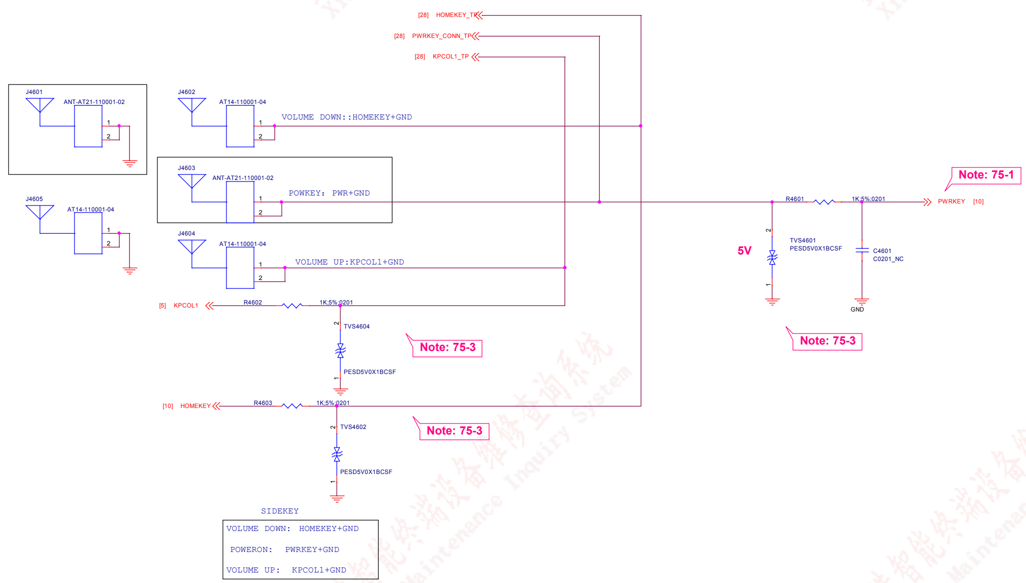

Accerometer
BMA253
U4412
SDO
SDX
VDDIO
NC
CSB
GND
GNDIO
VDD
SCX
PS
INT1
INT2
TP4401
TP4402
TP4403
TP4404
SPI4_MI
SPI4_MO
SPI4_CLK
SPI4_CSB
VIO18_PMU
VIO28_PMU
EINT_ACC
R4460
C4402
C4461
C4411
ALP-Sensor
U4416
GND1
INTB
SCL
SDA
VDD
NA4
GND2
NA3
NA2
NA1
MN26005UKDN
EINT_ALS
SCL1
SDA1
U4417
LEDA
LDR
/INT
NC
STK3332
VIO28_PMU_CON
VBAT
EINT_PS
FOR SAR sensor
U4450
IN
OUT
EN
GND1
GND2
R4461
C4452
C4453
C4450
C0201_NC
VDD2V8_SAR
ALS/PROXIMITY SENSOR
821400000421
I2C Address 0x47H
R4420
R4422
R4423
R4425
R0201_NC
R4424
C4404
C4410
C4405
C4406
Title 44_Sensor
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Monday, April 08, 2019
Sheet 25 of 53



Title	45_SIM/TF IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 26 of 53	

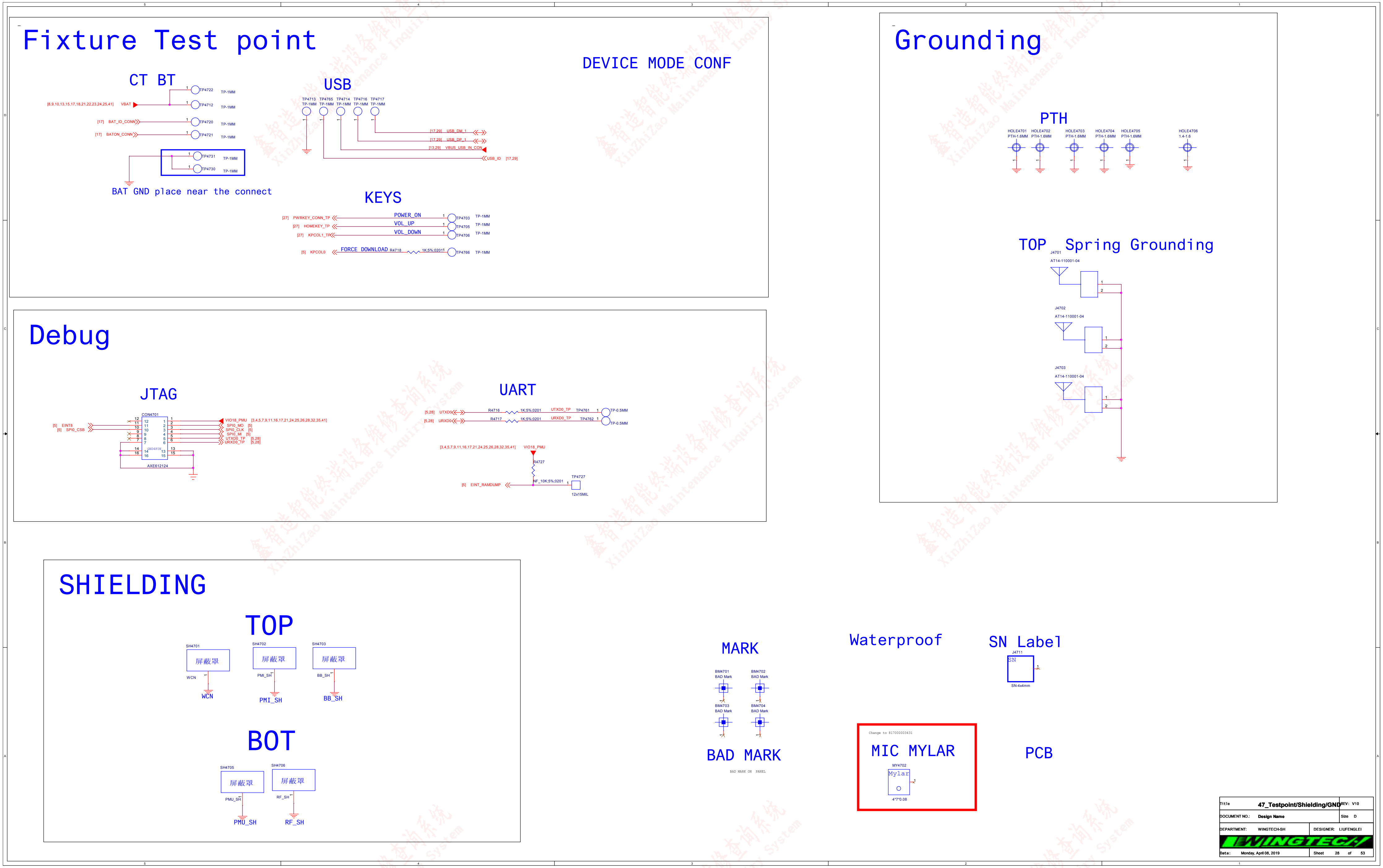
Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY

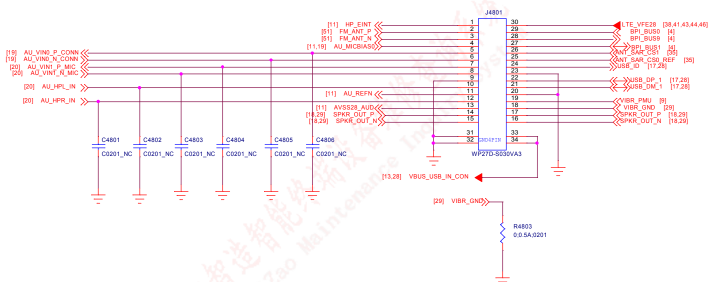


Note 75-1:DO NOT put pull-up resistor on PWRKEY
Note 75-2: Volume Up:HOME KEY+GND
Volume Down:KPCOL1+GND

Title	46_Sidekey	REV: V10	
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, April 11, 2019	Sheet 27	of 53



Fixture Test point
CT BT
USB
DEVICE MODE CONF
BAT GND place near the connect
KEYS
POWER_ON
VOL_UP
VOL_DOWN
FORCE_DOWNLOAD
Debug
JTAG
UART
Grounding
PTH
TOP Spring Grounding
SHIELDING
TOP
屏蔽罩
WCN
PMI_SH
BB_SH
BOT
PMU_SH
RF_SH
MARK
BAD MARK
Waterproof
SN Label
MIC MYLAR
PCB
Title 47_Testpoint/Shielding/GND
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
WINGTECH
Date: Monday, April 08, 2019
Sheet 28 of 53



Title	48_Sub PCB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI	
Date:	Monday, April 08, 2019	Sheet 29 of 53	

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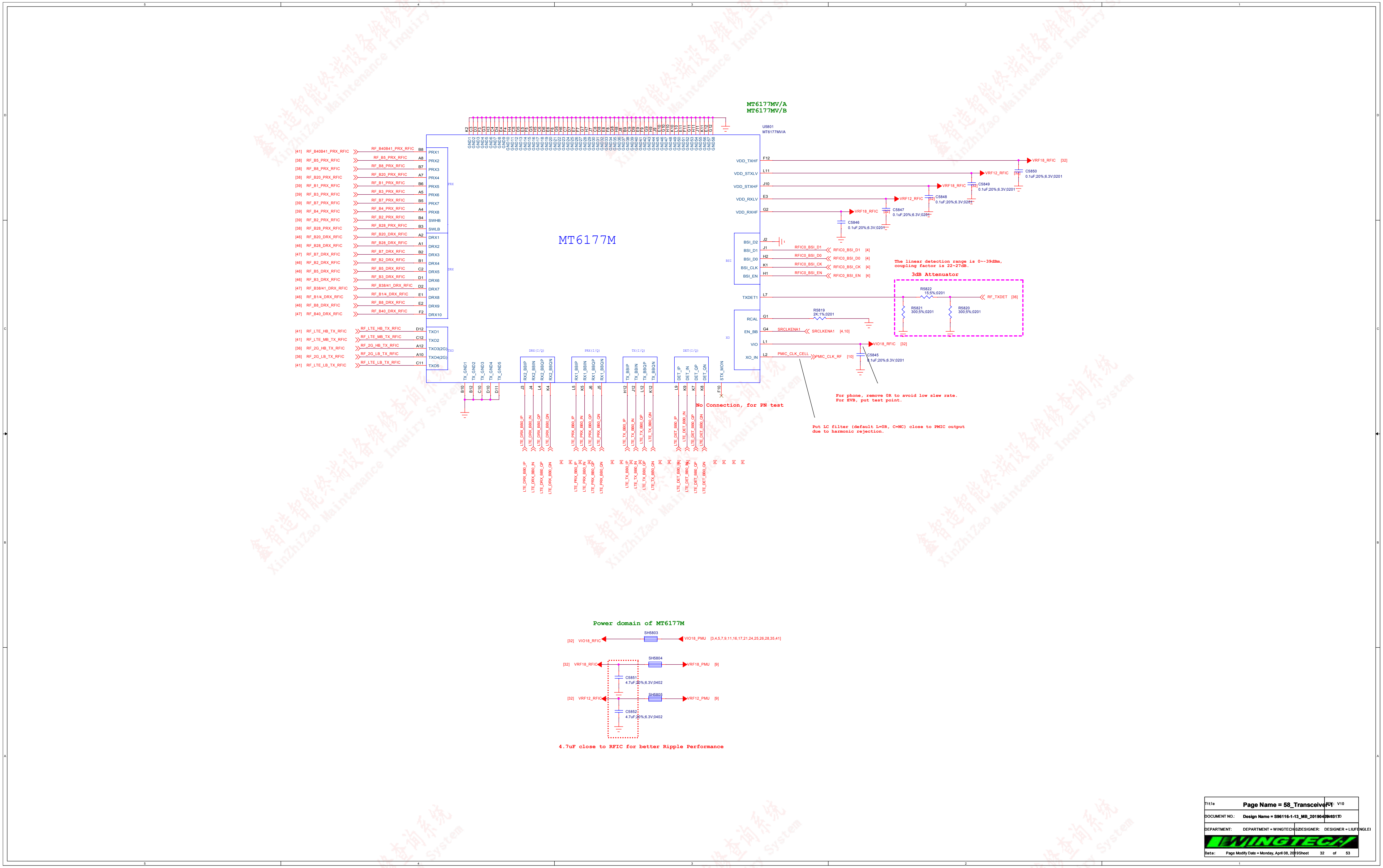
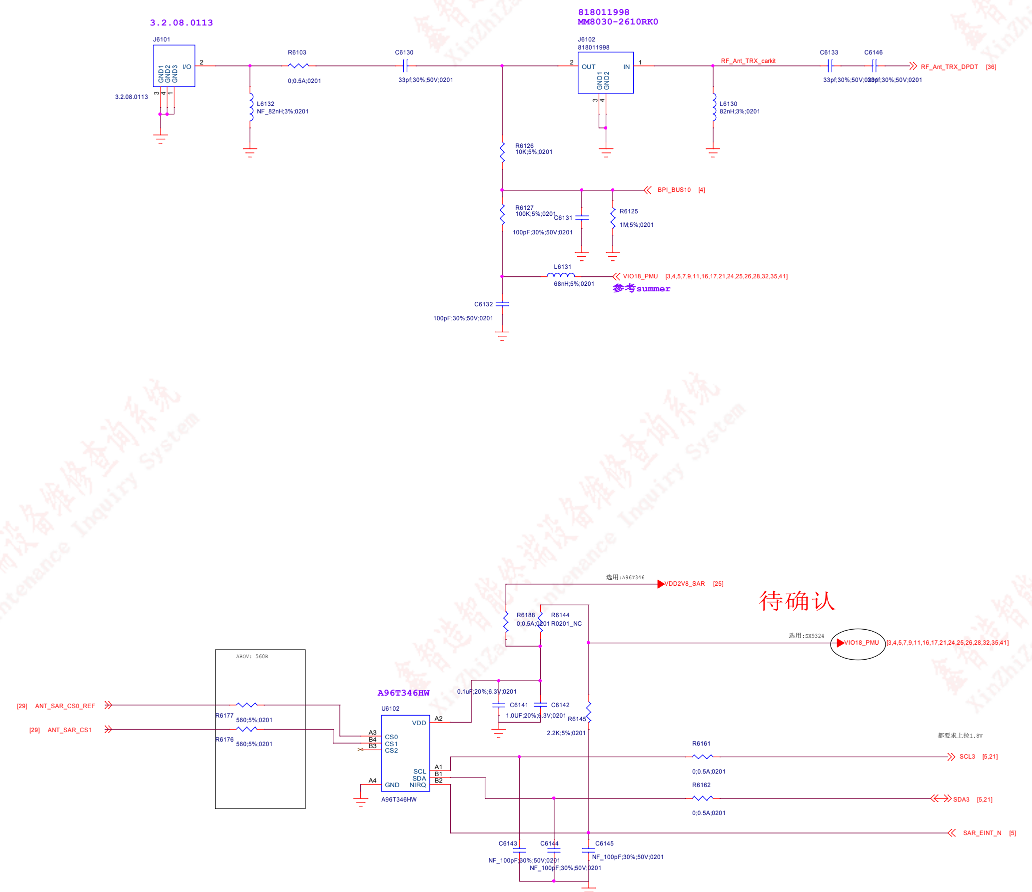
Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Monday, April 08, 2019	Sheet 31 of 53	



MT6177MV/A
MT6177MV/B
U5801
MT6177MV/A
MT6177M
VDD_TXHF
VDD_STXLV
VDD_STXHF
VDD_RXLV
VDD_RXHF
VRF18_RFIC
VRF12_RFIC
C5850
0.1uF;20%;6.3V;0201
C5849
C5848
C5847
C5846
BSI_D2
BSI_D1
BSI_D0
BSI_CLK
BSI_EN
RFIC0_BSI_D1
RFIC0_BSI_D0
RFIC0_BSI_CK
RFIC0_BSI_EN
TXDET1
RCAL
EN_BB
VIO
XO_IN
R5819
2K;1%;0201
SRCLKENA1
VIO18_RFIC
PMIC_CLK_CELL
PMIC_CLK_RF
C5845
0.1uF;20%;6.3V;0201
The linear detection range is 0~-39dBm, coupling factor is 22~27dB.
3dB Attenuator
R5822
15;5%;0201
R5821
300;5%;0201
R5820
300;5%;0201
RF_TXDET
For phone, remove 0R to avoid low slew rate.
For EVB, put test point.
Put LC filter (default L=0R, C=NC) close to PMIC output due to harmonic rejection.
No Connection, for PN test
RF_B40B41_PRX_RFIC
RF_B5_PRX_RFIC
RF_B8_PRX_RFIC
RF_B20_PRX_RFIC
RF_B1_PRX_RFIC
RF_B3_PRX_RFIC
RF_B7_PRX_RFIC
RF_B4_PRX_RFIC
RF_B2_PRX_RFIC
RF_B28_PRX_RFIC
RF_B20_DRX_RFIC
RF_B28_DRX_RFIC
RF_B7_DRX_RFIC
RF_B2_DRX_RFIC
RF_B5_DRX_RFIC
RF_B3_DRX_RFIC
RF_B38/41_DRX_RFIC
RF_B1/4_DRX_RFIC
RF_B8_DRX_RFIC
RF_B40_DRX_RFIC
RF_LTE_HB_TX_RFIC
RF_LTE_MB_TX_RFIC
RF_2G_HB_TX_RFIC
RF_2G_LB_TX_RFIC
RF_LTE_LB_TX_RFIC
PRX1
PRX2
PRX3
PRX4
PRX5
PRX6
PRX7
PRX8
SWHB
SWLB
DRX1
DRX2
DRX3
DRX4
DRX5
DRX6
DRX7
DRX8
DRX9
DRX10
TXO1
TXO2
TXO3(2G)
TXO4(2G)
TXO5
LTE_DRX_BB0_IP
LTE_DRX_BB0_IN
LTE_DRX_BB0_QP
LTE_DRX_BB0_QN
LTE_PRX_BB0_IP
LTE_PRX_BB0_IN
LTE_PRX_BB0_QP
LTE_PRX_BB0_QN
LTE_TX_BB0_IP
LTE_TX_BB0_IN
LTE_TX_BB0_QP
LTE_TX_BB0_QN
LTE_DET_BB0_IP
LTE_DET_BB0_IN
LTE_DET_BB0_QP
LTE_DET_BB0_QN
Power domain of MT6177M
SH5803
VIO18_RFIC
VIO18_PMU
SH5804
VRF18_RFIC
VRF18_PMU
SH5805
VRF12_RFIC
VRF12_PMU
C5851
4.7uF;20%;6.3V;0402
C5852
4.7uF;20%;6.3V;0402
4.7uF close to RFIC for better Ripple Performance
Page Name = 58_Transceiver1
V10
Design Name = S96116-1-13_MB_20190408
DEPARTMENT = WINGTECH
DESIGNER = LIUFENGLEI
WINGTECH
Page Modify Date = Monday, April 08, 2019
Sheet 32 of 53

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 33 of 53	

Title	Reserved	REV: V10
DOCUMENT NO.:	Design Name	Size A1
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019		Sheet 34 of 53



Title	REV: V10
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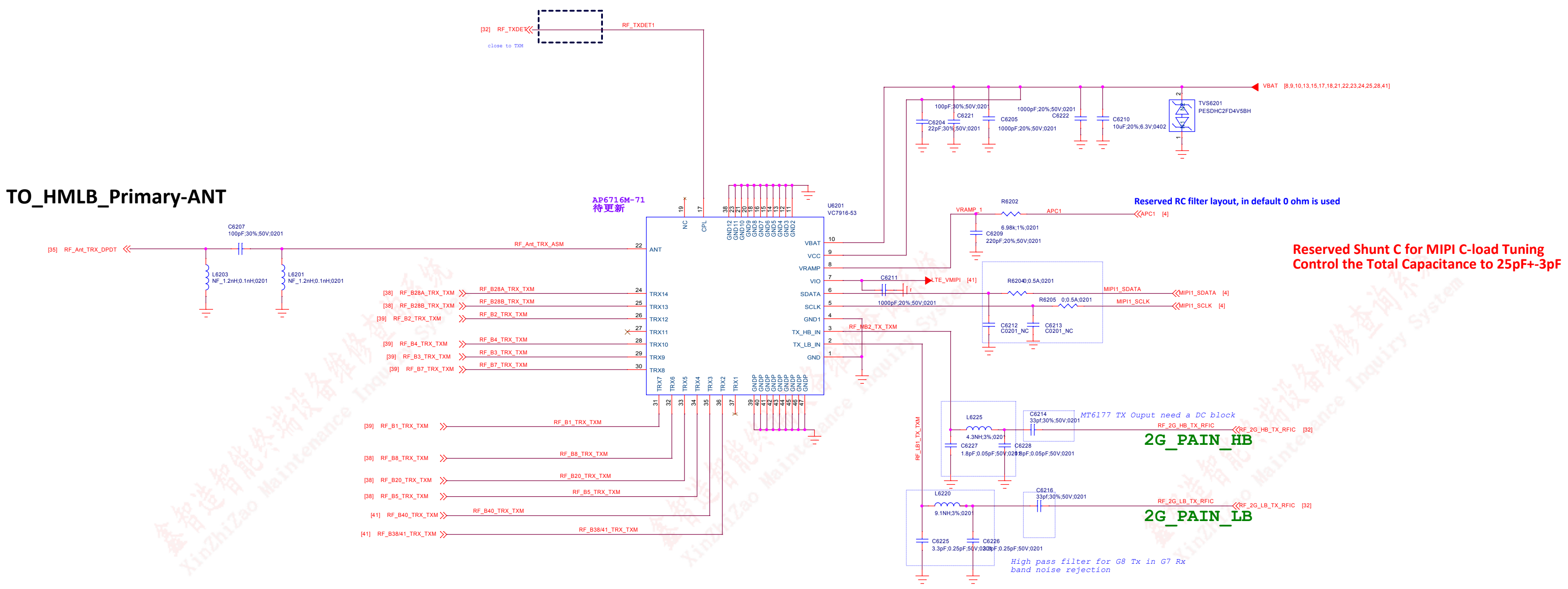
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date: Page Modify Date = Monday, April 08, 2019	Sheet 35 of 53



Title	Page Name = 62_Primary_ASM	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190408-0317	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019	Sheet 36 of 53

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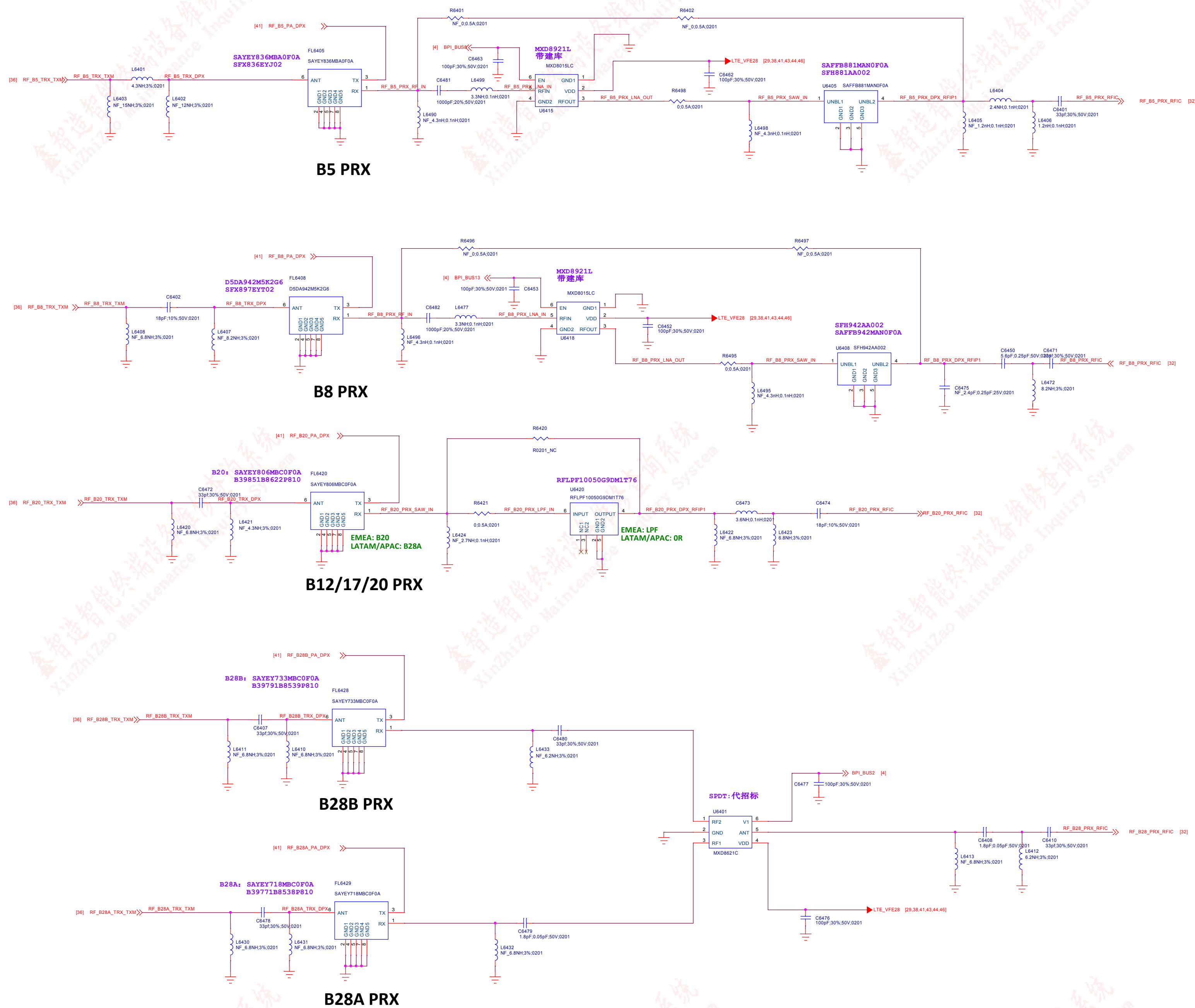
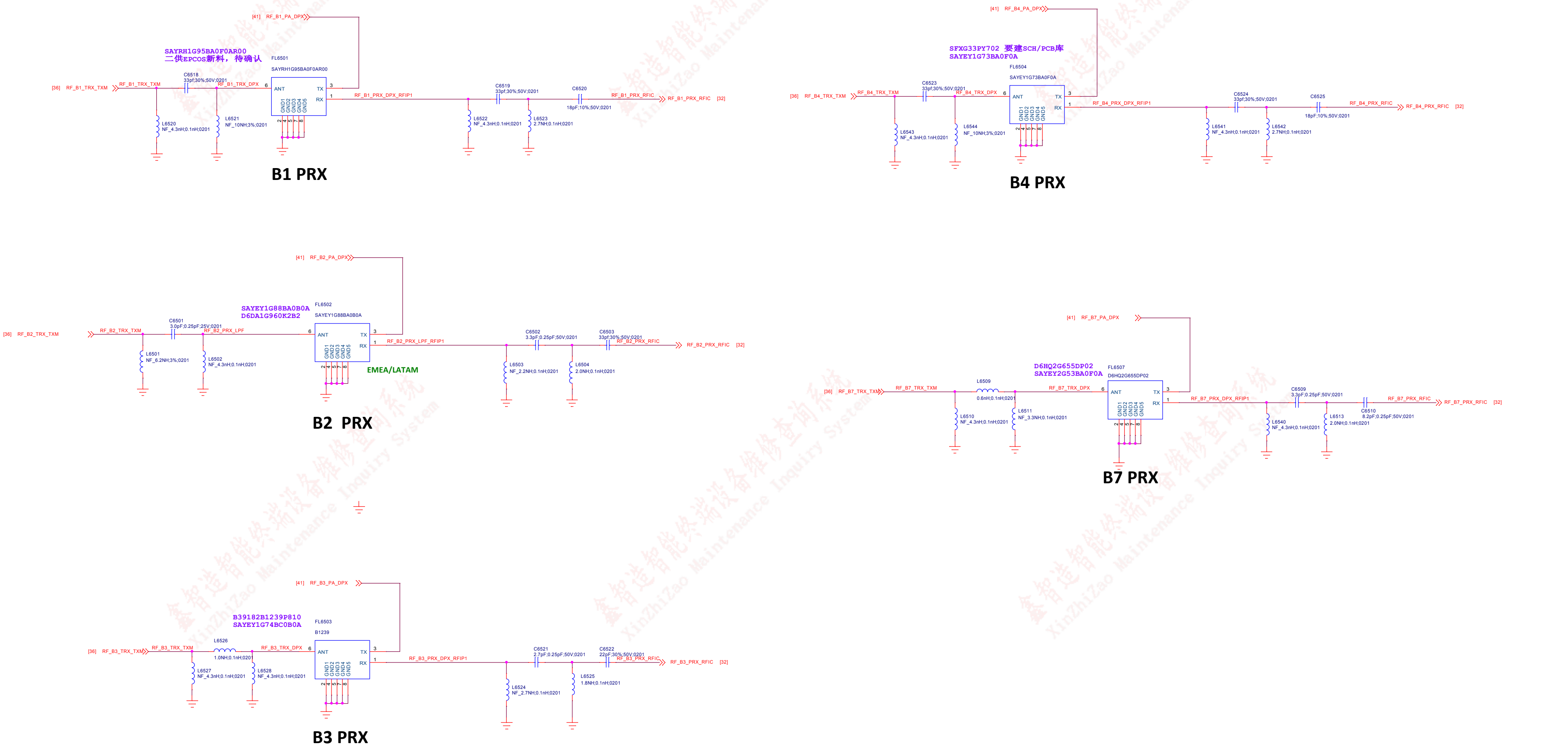
Title	<Title>				
Size A1	Document Number <Doc>				Rev 1.0
Date:	Monday, April 08, 2019	Sheet	37	of	53

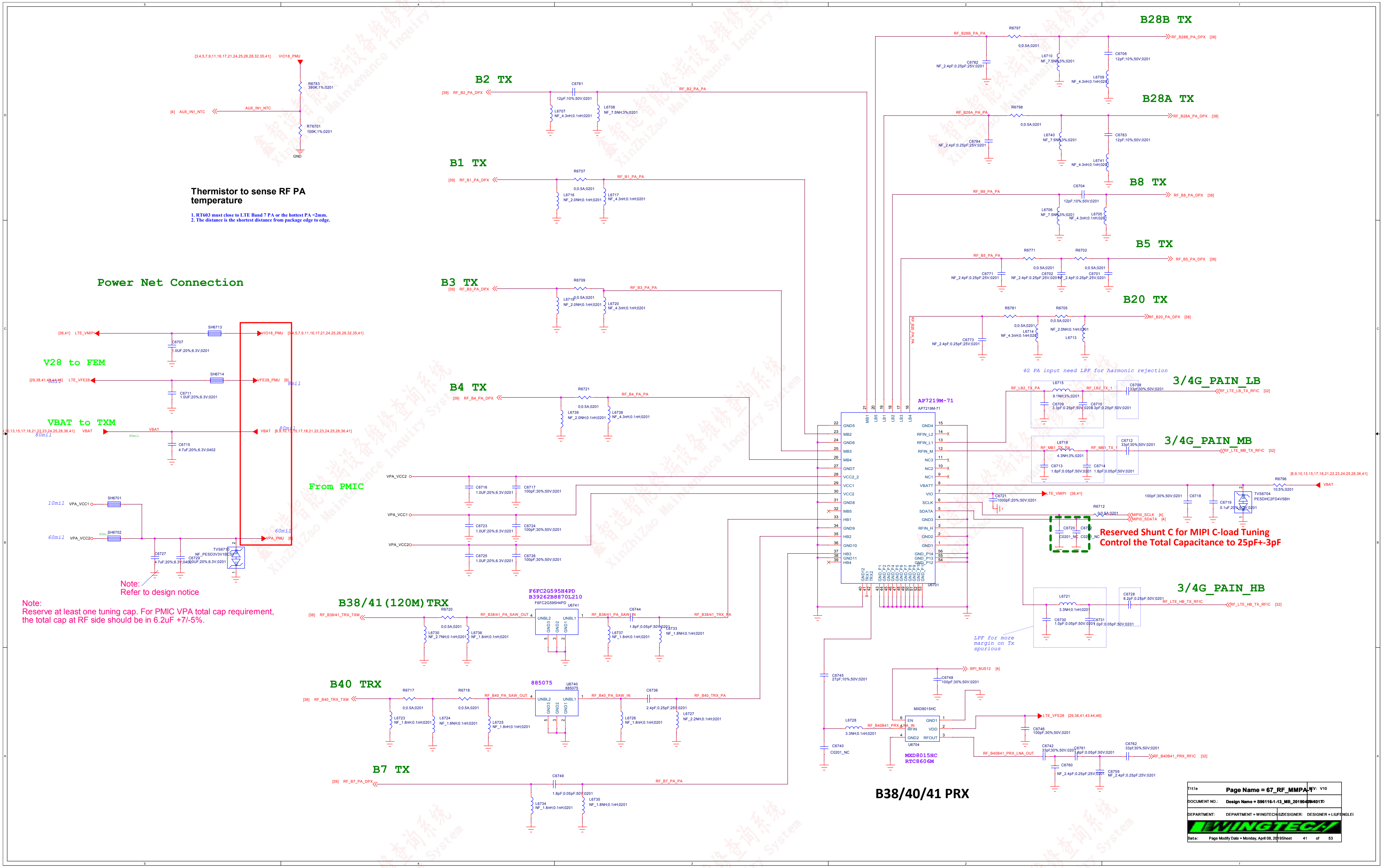


Title	Page Name = 64_TRX_LB	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317	Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019	Sheet 38 of 53



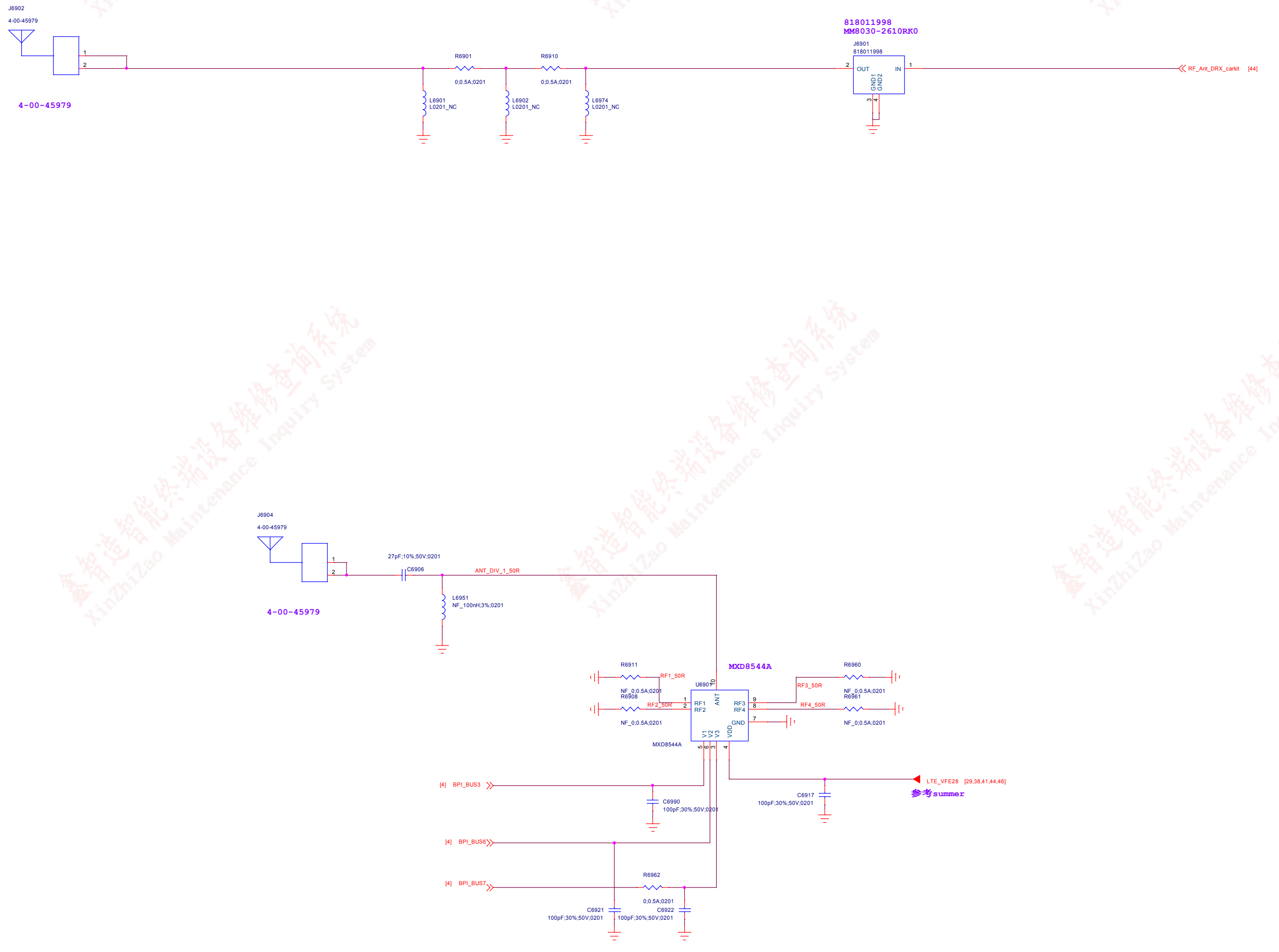
Title	Page Name = 65_TRX_MHB	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190408-1517	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019	Sheet 39 of 53

Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Monday, April 08, 2019	Sheet 40 of 53	

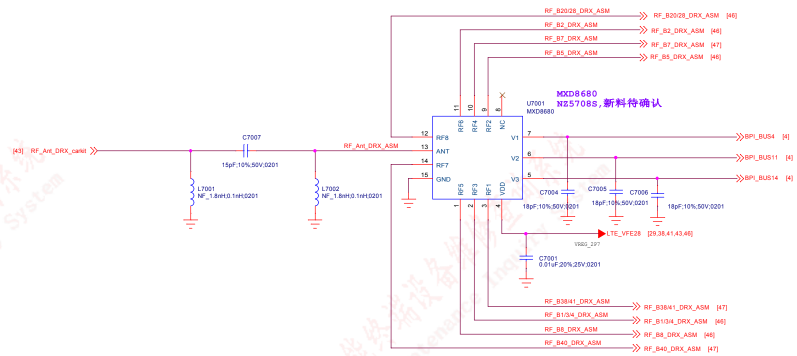


B28B TX
B28A TX
B2 TX
B1 TX
B8 TX
B5 TX
B3 TX
B20 TX
Thermistor to sense RF PA temperature
1. RT603 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.
Power Net Connection
V28 to FEM
VBAT to TXM
B4 TX
From PMIC
4G PA input need LPF for harmonic rejection
3/4G_PAIN_LB
3/4G_PAIN_MB
AP7219M-71
Reserved Shunt C for MIPI C-load Tuning
Control the Total Capacitance to 25pF+-3pF
3/4G_PAIN_HB
Note:
Refer to design notice
Note:
Reserve at least one tuning cap. For PMIC VPA total cap requirement, the total cap at RF side should be in 6.2uF +7/-5%.
B38/41(120M)TRX
F6FC2G595H4PD
B39262B8870L210
LPF for more margin on Tx spurious
B40 TRX
885075
MXD8015HC
RTC8606M
B7 TX
B38/40/41 PRX
Page Name = 67_RF_MMPA
Design Name = S96116-1-13_MB_20190409
DEPARTMENT = WINGTECH
WINGTECH
Page Modify Date = Monday, April 08, 2019
Sheet 41 of 53

Title	<Title>			
Size A1	Document Number <Doc>			Rev 1.0
Date:	Monday, April 08, 2019	Sheet	42	of 53

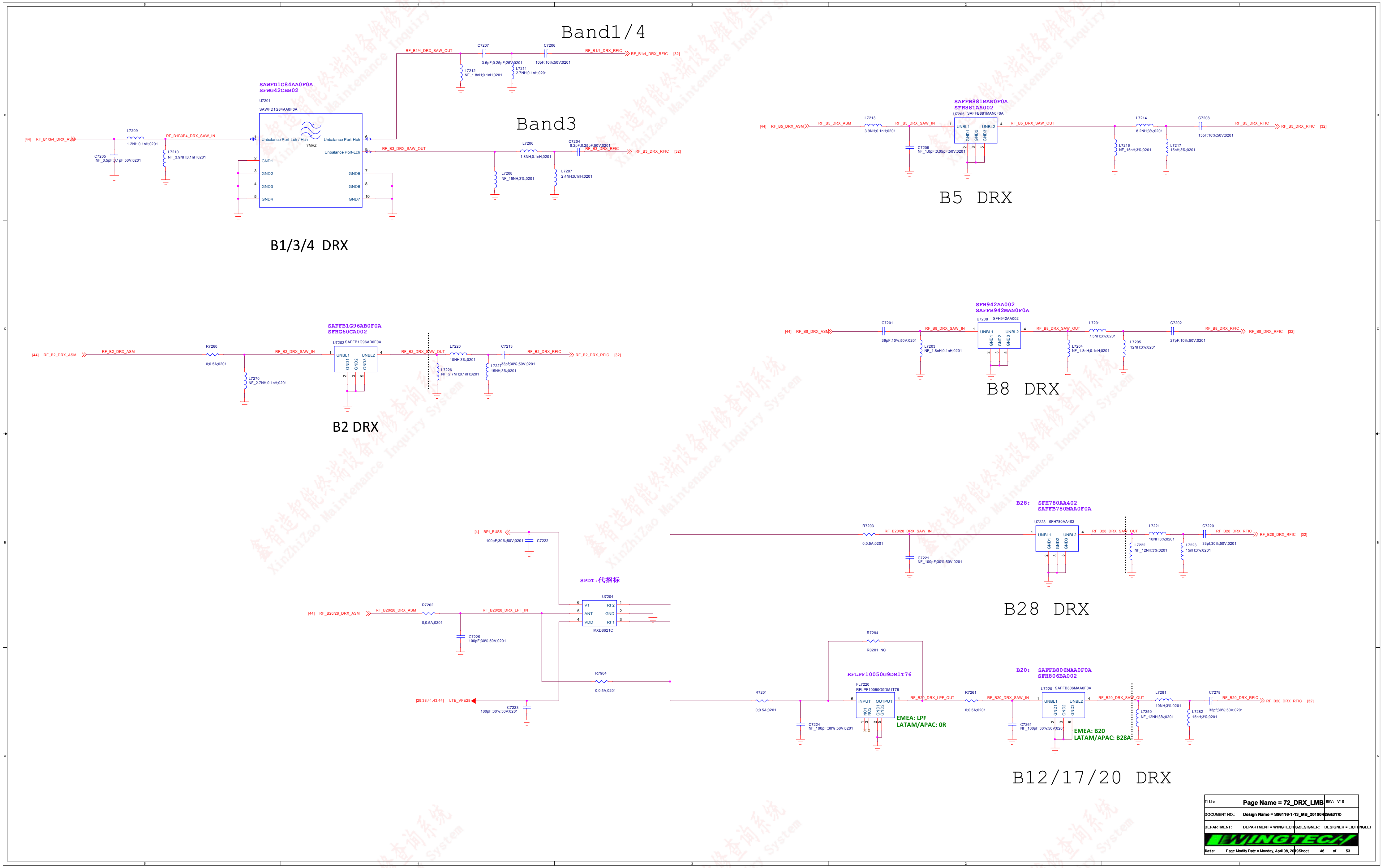


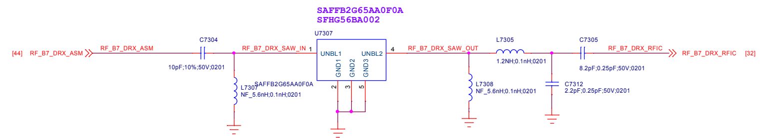
Title	Page Name = 69_Diversity_ANT	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1337	Size D
DEPARTMENT:DEPARTMENT = WINGTECH-SZ	DESIGNER:DESIGNER = LIUFENGLEI	
Date:Page Modify Date = Monday, April 08, 2019	Sheet 43 of 53	



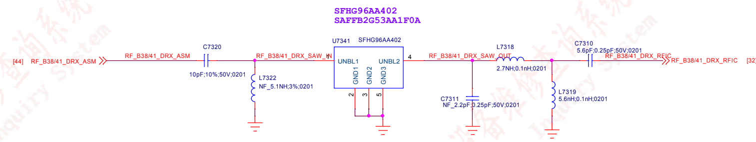
Title	Page Name = 70_Diversity ASM	REV	V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317	Size	D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019	Sheet 44	of 53

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 45	of 53

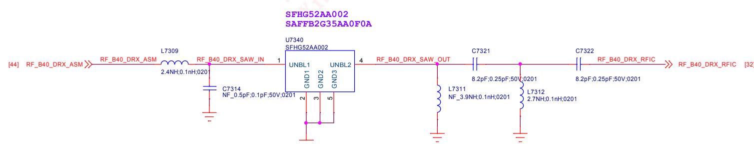

Band1/4
Band3
B1/3/4 DRX
SAWFD1G84AA0F0A
SFWG42CBB02
B5 DRX
SAFFB881MAN0F0A
SFH881AA002
B2 DRX
SAFFB1G96AB0F0A
SFHG60CA002
B8 DRX
SFH942AA002
SAFFB942MAN0F0A
B28 DRX
B28: SFH780AA402
SAFFB780MAA0F0A
SPDT:代招标
RFLPF10050G9DM1T76
EMEA: LPF
LATAM/APAC: 0R
B20: SAFFB806MAA0F0A
SFH806BA002
EMEA: B20
LATAM/APAC: B28A
B12/17/20 DRX
Page Name = 72_DRX_LMB
REV: V10
Design Name = S96116-1-13_MB_20190408
DEPARTMENT = WINGTECH SZ
DESIGNER = LIUFENGLEI
WINGTECH
Page Modify Date = Monday, April 08, 2019
Sheet 46 of 53



B7 DRX



B38/41 DRX



B40 DRX

Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
WINGTECH		
Date: Page Modify Date = Monday, April 08, 2019	Sheet 47 of 53	

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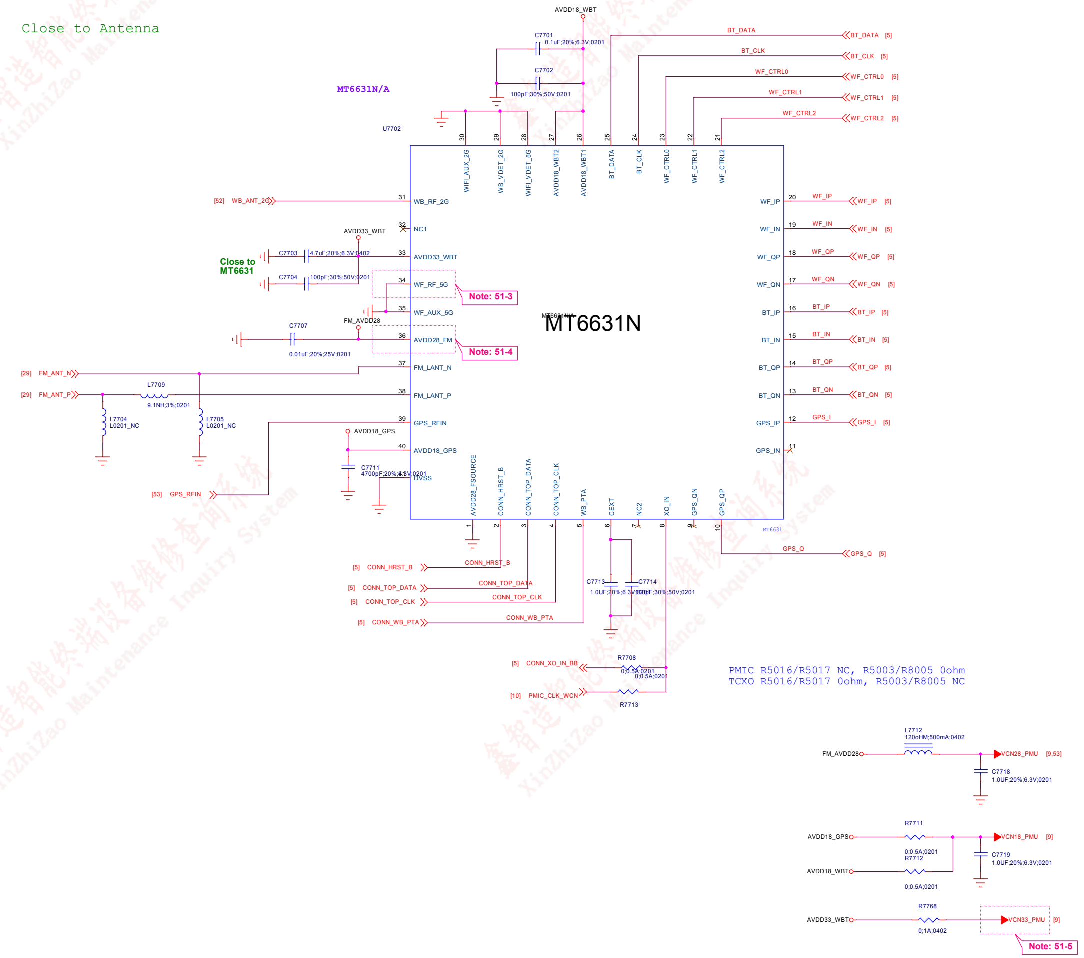
Title	<Title>		
Size A1	Document Number <Doc>		Rev 1.0
Date: Monday, April 08, 2019		Sheet 48 of 53	

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 49	of 53

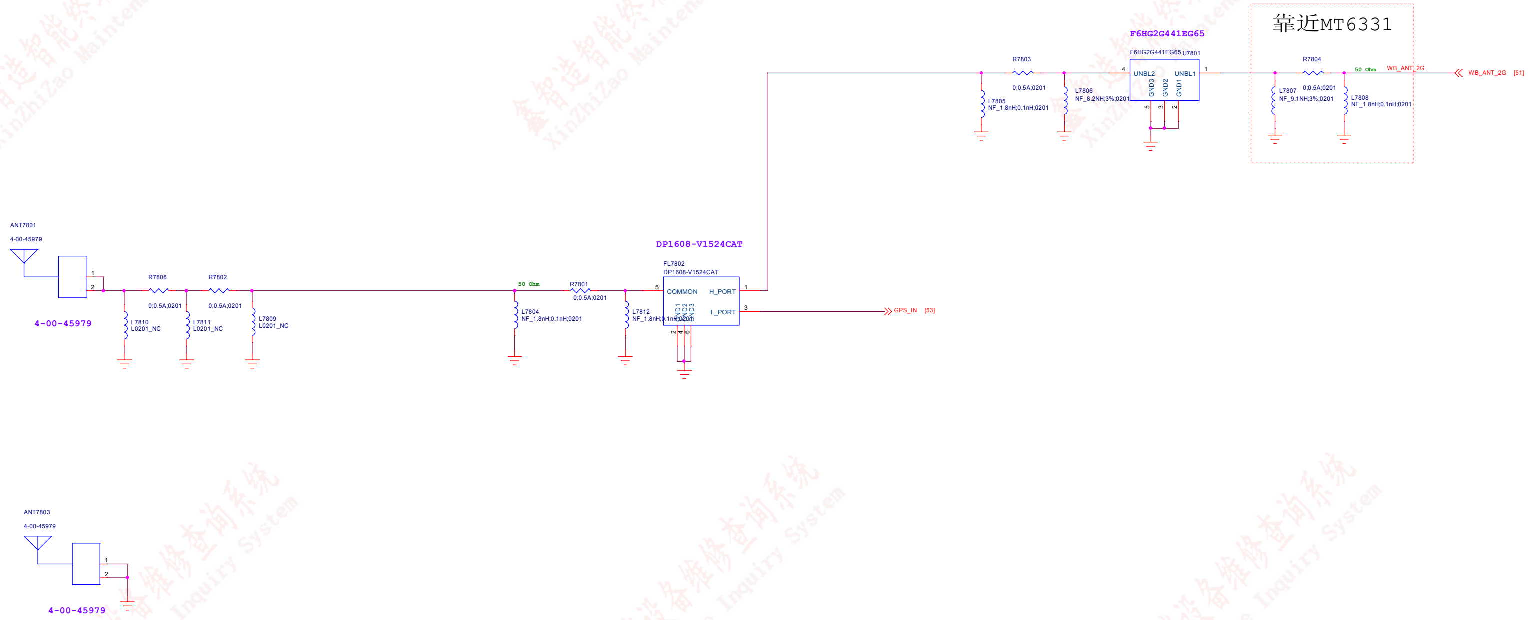
Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 50 of 53	

Close to Antenna

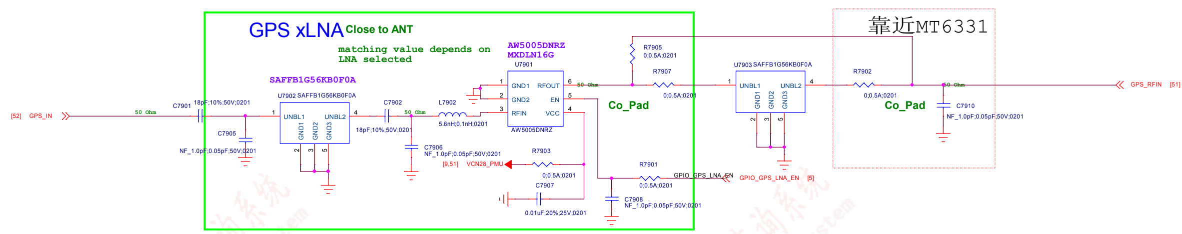


Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"	
Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6762 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support
Note 51-5:	If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)

Title	Page Name = 77_WCN_MT6631 Rev: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317 Size: D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019 Sheet 51 of 53



Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 52 of 53	



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DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 53	of 53